Chapter 9

Banks' balance sheet and income structure

Learning objectives

- To understand the importance of banks' financial statements
- To identify the main assets and liabilities of commercial and investment banks
- To understand the sources of revenue for commercial and investment banks
- To understand the importance of economic capital
- To describe the concept of shareholder value creation and the cost of equity capital
- To become familiar with the most commonly used bank financial ratios
- To understand the growing importance of non-financial performance

9.1 Introduction

Traditionally, the business of banks is to intermediate funds between surplus units and deficit units thereby linking depositors with borrowers. Banks also provide pooling of risk, liquidity services and undertake delegated monitoring. Financial intermediaries can be classified according to their different **balance sheet** structures. For deposit-taking institutions, the main source of funding (customer deposits) are reported on the liabilities side of the balance sheet, while the allocation of these funds (cash, loans, investments and fixed assets) is detailed on the assets side. Banks' profits are derived from the **income statement (profit and loss account)**, a document that reports data on costs and revenues and measures bank performance over two balance sheet periods. This chapter focuses on understanding commercial and investment banks' financial statements and describes the main characteristics of their balance sheet and income statements. The last part of the chapter investigates the most common bank financial ratios such as **return-on-assets**, **return-on-equity**, **net interest margin** and the **cost–income ratio**. It also discusses the growing importance of banks' **non-financial performance**, particularly in relation to the management of social and environmental challenges.

9.2 Retail banks' balance sheet structure

The **balance sheet** is a financial statement of the wealth of a business or other organisation on a given date. This is usually at the end of the financial year. For commercial banks the balance sheet (also known as *Report of Condition* in the US) lists all the stock values of sources and uses of banks' funds. Banks' funds come from:

(a) the general public (retail deposits);

(b) companies (small, medium and large corporate deposits);

(c) other banks (interbank deposits);

(d) equity issues (share issues, conferring ownership rights on holders);

(e) debt issues (bond issues and loans); and

(f) saving past profits (retained earnings).

The above is generally classified as banks' **liabilities** (debt) and **capital** (equity). These funds are then transformed into financial and, to a lesser extent, real **assets**:

(a) cash;

(b) liquid assets (securities);

(c) short-term money market instruments such as Treasury bills, which banks can sell (liquidate) quickly if they have a cash shortage;

(d) loans;

(e) other investments; and

(f) fixed assets (branch network, computers, premises).

Table 9.1 summarises the assets and liabilities in a simplified commercial bank balance sheet.

Bank liabilities (e.g. retail deposits) tend to have shorter maturities than assets (e.g. mortgage loans). This mismatch derives from the different requirements of depositors and borrowers: typically, the majority of depositors want to lend their assets for short periods of time and for the highest possible return. In contrast, the majority of borrowers require loans that are cheap and for long periods. The asset transformation function of banks is derived from these characteristics. To recap, banks have the primary function of being asset transformers because they intermediate between depositors and borrowers by changing the characteristics of their liabilities as they move from one side of the balance sheet to the other.

Table 9.1 Simplified commercial bank balance sheet

Assets	Liabilities
Cash	Deposits: retail
Liquid assets	Deposits: wholesale
Loans	
Other investments	
Fixed assets	
	Equity
	Other capital terms
Total assets	**Total liabilities and equity**

Capital (see also Section 9.2.1.3) is sometimes referred to as equity capital or net worth and is equal to the difference between assets and liabilities.

9.2.1 Assets and liabilities of commercial banks: main components

The balance sheet provides information about the bank's financial position at the end of the accounting period. It comprises three principal components: (a) the assets the bank controls; (b) the liabilities the bank is obliged to meet; and (c) the equity interests of the bank's owners.

Tables 9.2 and 9.3 exhibit the combined balance sheet for UK banks as reported by the Bank of England. The tables show aggregate assets and liabilities of all financial institutions recognised by the Bank of England as UK banks for statistical purposes.

9.2.1.1 The assets side

On the asset side, banks store a relatively small amount (about 0.3 per cent of total assets in 2019) of cash in the form of *notes and coins* to meet daily commitments. In the United Kingdom, according to the current regulation, both banks and building societies with average eligible liabilities of £600 million or more are required to hold non-operational, non-interest-bearing deposits with the Bank of England of 0.18 per cent. The purpose of these deposits (known as *cash ratio deposits*) is to ensure banks' liquidity. Banks can also keep *other balances* with the Bank of England (i.e. other than cash ratio deposit); these deposits give the central bank a source of income.

In case of cash shortage, banks can ask for a loan in the interbank market. The interbank market constitutes an important portion of the money markets and it is the place where banks meet each day to exchange liquidity. Therefore, the item *market loans* in the asset side of a bank balance sheet includes wholesale loans that are typically very short-term (i.e. overnight or 'call' loans), very liquid (they allow banks to lend money and call them back at short notice) and characterised by large volumes (typically >£1 million).

Bankers' acceptances are negotiable time drafts, or bills of exchange, that have been accepted by a bank that, by accepting, assumes the obligation to pay the holder of the draft the face amount of the instrument on the maturity date specified. They are used primarily to finance the export, import, shipment or storage of goods. *Acceptances granted* comprise a claim on the party whose bill the banks have accepted, except for bills both accepted and discounted by the same bank that are included as lending (unless subsequently rediscounted).

Another important source of liquidity is provided by *bills*. As shown in Table 9.2, the main bills held by UK banks are *Treasury bills* (or *T-bills*), which are essentially a form of short-term government borrowing; *bank bills* (usually eligible for rediscounting at the Bank of England) and other short-term bills including *local government bills* and *public corporation bills*.

Further liquidity is provided by the item claims under *sale and repurchase agreements*. This item comprises cash claims arising from the purchase of securities for a finite period with a commitment to re-sell.

By far the most important item on the asset side, *advances*, includes all balances with, and lending to, customers not included elsewhere. Despite the dramatic changes that have characterised the banking sector in recent years, loans are still the primary earning assets of banks, and account for a relatively large proportion of total assets. As reported in Table 9.2, in 2019 loans were the largest items on the balance sheet: sterling advances held on the asset side of banks in the United Kingdom totalled over £2.1 trillion, which was over 55 per cent of total sterling assets. Typically, UK banks lend to individuals, financial and non-financial firms. The

Table 9.2 Bank of England aggregate assets of UK banks (end-year 2019, £bn amounts)

	£bn end-year 2019	% over total sterling assets
Notes and coin	11.1	0.28%
With UK central bank	**455.6**	**11.44%**
– Cash ratio deposit	8.8	0.22%
– Other	446.8	11.22%
Market loans	**373.1**	**9.37%**
– UK banks	277.16	6.96%
– UK banks' CDs and commercial paper	2.41	0.06%
– Non-residents	93.52	2.35%
Acceptances granted	**0.2**	**0.00%**
– UK banks	–	–
– UK public sector	–	–
– Other UK residents	0.14	0.00%
– Non-residents	0.03	0.00%
Bills	**13.2**	**0.33%**
– Treasury bills	14.14	0.35%
– UK bank bills	0.00	0.00%
– Other UK	0.51	0.01%
– Non-residents	0.24	0.01%
Sales and repurchase agreements	**431.6**	**10.84%**
– UK banks	31.09	0.78%
– UK public sector	0.41	0.01%
– Other UK residents	276.82	6.95%
– Non-residents	123.32	3.10%
Advances	**2,195.39**	**55.12%**
– UK public sector	7.16	0.18%
– Other UK residents	2,092.03	52.52%
– Non-residents	96.19	2.42%
Investments	**437.5**	**10.98%**
– UK government bonds	115.76	2.91%
– Other UK public sector	0.45	0.01%
– UK banks	46.73	1.17%
– Other UK residents	200.61	5.04%
– Non-residents	73.96	1.86%
Items in suspense and collection	18.9	0.47%
Accrued amounts receivable	15.6	0.39%
Other assets	29.3	0.74%
Total sterling assets	**3,983.2**	**100%**
Total foreign currency assets	4,002.1	
– Of which total euro assets	1,422.9	
Total assets	**7,985.2**	

Note: Figures may not add due to rounding.

Source: Bank of England, Monetary and Financial Statistics interactive database, Table B1.4, **www.bankofengland.co.uk/statistics/tables**, and authors' calculations.

major categories of loans are *commercial loans* (such as short-term loans to businesses), *consumer loans* (for example, overdrafts and credit card loans), *mortgage lending* and *real estate loans* (such as long-term loans to finance commercial real estate such as office buildings).

The next item on the asset side of the balance sheet is *investments*. These include all longer-term securities beneficially owned by the reporting institution and include securities that the reporting institution has sold for a finite period, but with a commitment to repurchase (i.e. repos), but exclude securities that have been bought for a finite period, but with a commitment to resell (i.e. reverse repos). Securities are defined as marketable or potentially marketable income-yielding instruments including bonds, floating rate notes (FRNs), preference shares and other debt instruments, but excluding certificates of deposit and commercial paper that are shown as market loans.

The remaining assets include the following:

- *Items in suspense and collection*, which include, for example, debit balances awaiting transfer to customers' accounts and balances awaiting settlement of securities transactions. Collections comprise cheques drawn, and in course of collection, on other UK banks and building societies.
- *Accrued amounts receivable* are gross amounts receivable, but have not yet been received, and include interest and other revenues.
- *Other assets* include holdings of gold bullion and gold coin, other commodities, together with land, premises, plant and equipment and other physical assets owned, or recorded as such, including assets leased out under operating leases. Assets leased out under finance leases are included as loans.
- *Eligible banks' total sterling acceptances* comprise all bills accepted by a reporting institution whose bills are eligible for rediscount at the Bank of England including those that the reporting institution has itself discounted.

Finally, in 2019, UK banks had about £4,002 billion in foreign currency assets (e.g. foreign currency loans), of which approximately 36 per cent were euro-denominated. As shown in Table 9.2, foreign currency assets and liabilities account for a significant proportion of total bank assets.

9.2.1.2 The liability side

On the liabilities side, as illustrated in Table 9.3, the first item reported is *notes outstanding and cash-loaded cards*. This includes all notes and cash held by banks, including the sterling notes issued by Scottish and Northern Irish banks and cash-loaded cards issued by banks (these are electronic cards, smart cards, etc.).

The largest proportion of bank liabilities is in the form of deposits that are typically made by individuals and firms, including deposits by other UK banks. The majority of deposits are represented by sight and time deposits (55 and 31 per cent, respectively), as shown in Figure 9.1.

Sight deposits comprise those deposits where the entire balance is accessible without penalty, either on demand or by close of business on the day following the one on which the deposit was made. *Time deposits* comprise all other deposits and they include, for example, 30- and 60-day savings bank deposits and ISA deposits.[1]

[1] ISAs (Individual Savings Accounts) were introduced in the United Kingdom in 1999. These are tax-free savings and investment accounts that can be used to save cash, or invest in stocks and shares.

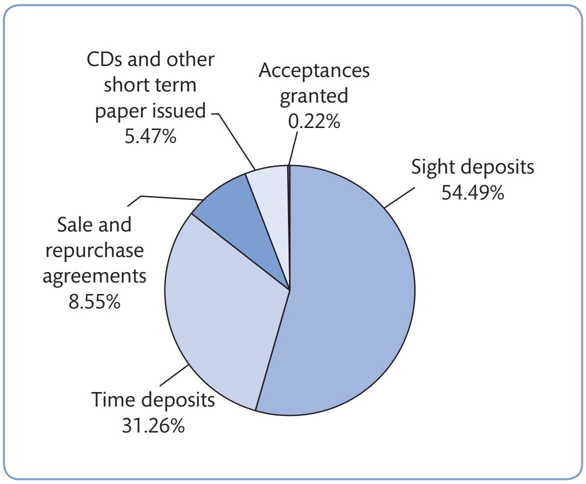

Figure 9.1 **Breakdown of UK banks' sterling deposits, end-year 2019**

Source: Bank of England, Monetary and Financial Statistics Interactive Database and authors' calculations.

As shown in Figure 9.1, deposits encompass all credit balances on customers' accounts, including acceptances granted, liabilities under sale and repurchase agreements and certificates of deposit.

Acceptances granted represent the banks' liabilities to the owners of bills. *Liabilities under sale and repurchase agreements* comprise cash receipts arising from the sale of securities or other assets that the bank has sold temporarily with a commitment to repurchase.

CDs and other short-term paper issued: certificates of deposits (CDs) are certificates given to depositors in return for a (wholesale) deposit. The holder of the CD receives interest at a fixed or floating rate. CDs are short-term securities and are re-saleable in the market. This item also contains promissory notes issued by the reporting institutions, unsubordinated capital market instruments (except debentures and secured loan stocks) of any maturity and subordinated loan stocks with a maturity of five years or less. Other subordinated loan stocks and debentures are included in capital and other funds (see below for the details on the capital item).

The remaining non-deposit liabilities, as reported in Table 9.3, include the following:

- *Items in suspense and transmission*, such as balances awaiting settlement of securities transactions, standing orders and credit transfers debited to customers' accounts, and other items for which the corresponding payment has not yet been made by the reporting institution.
- *Net derivatives*, which comprise the overall net derivatives position of contracts that are included within the trading and banking books of the reporting institutions.
- *Accrued amounts payable*, which are gross amounts payable that have not yet been paid or credited to accounts.
- *Capital and other internal funds*, which consist primarily of shareholders' funds, reserves and long-term debt.

Finally, Table 9.3 shows that in 2019 UK banks had over £4 trillion in foreign currency liabilities (e.g. foreign currency sight and time deposits), of which approximately 42 per cent were euro-denominated.

Table 9.3 Bank of England aggregate liabilities of UK banks (end-year 2019, £bn amounts)

Liabilities	£bn end-year 2019	% over total sterling assets
Notes outstanding and cash-loaded cards	**7.7**	**0.20%**
Sight deposits	**1,867.12**	**47.9%**
– UK banks	156.4	4.0%
– UK public sector	18.7	0.5%
– Other UK residents	1,512.6	38.8%
– Non-residents	179.4	4.6%
Time deposits	**1,071.09**	**27.5%**
– UK banks	230.9	5.9%
– UK public sector	14.7	0.4%
– Other UK residents	669.0	17.2%
– Non-residents	156.5	4.0%
Sale and repurchase agreements	**293.1**	**7.5%**
– UK banks	39.8	1.0%
– UK public sector	0.1	0.0%
– Other UK residents	141.6	3.6%
– Non-residents	111.5	2.9%
Acceptances granted	**0.2**	**0.00%**
CDs and other short term paper issued	**187.5**	**4.8%**
Total sterling deposits	**3,418.90**	**87.8%**
Items in suspense and transmission	17.5	0.4%
Net derivatives	4.1	0.1%
Accrued amounts payable	14.8	0.4%
Capital and other internal funds	431.3	11.1%
Total sterling liabilities	**3,894.30**	**100.0%**
Total foreign currency liabilities	4,090.93	
– Of which total euro liabilities	1,446.16	
TOTAL LIABILITIES	7,985.2	

Source: Bank of England, Monetary and Financial Statistics Interactive database, Table B1.4, **www.bankofengland.co.uk/statistics/tables** and authors' calculations.

The assets and liabilities side of a major UK bank

Tables 9.4 and 9.5 illustrate the consolidated financial data over 2013–2017 for a major UK credit institution, Barclays Bank. Figures 9.2 and 9.3 illustrate the breakdown of the major components of Barclays Bank's assets and liabilities in 2017.

A few remarks about Barclays' recent past are necessary before commenting on these financial results. Due to the ringfencing rules that apply to all large UK banks with a 3-year average of more than £25 billion 'core deposits' (in 2018 Barclays had to isolate core retail banking operations from the rest of the business (i.e. investment banking and international banking). 'Barclays Bank UK PLC' officially became the first UK bank to deliver a ring-fenced banking structure, which includes personal and business banking and Barclaycard consumer UK businesses. New governance arrangements mean that the bank has its own separately

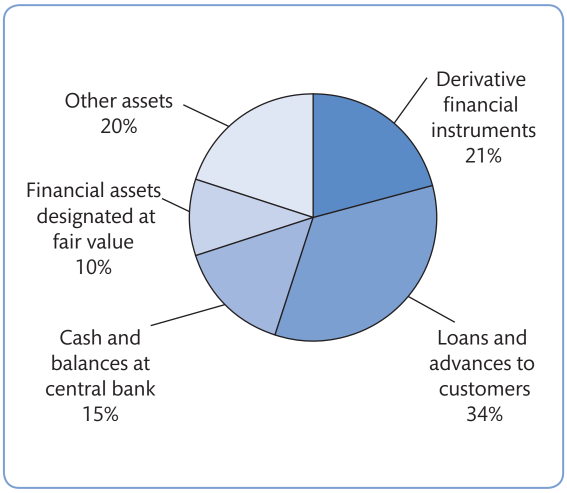

Figure 9.2 **Barclays Bank main assets (%), end-year 2017**

Source: Data from **www.investorrelations.barclays.co.uk** and authors' calculations.

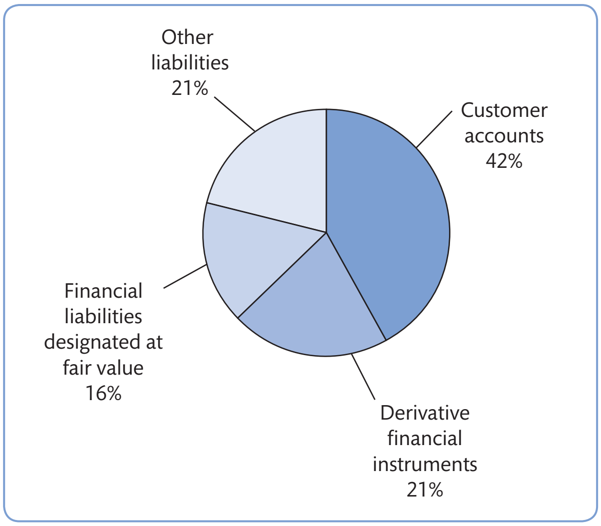

Figure 9.3 **Barclays Bank main liabilities (%), end-year 2017**

Source: Data from **www.investorrelations.barclays.co.uk** and authors' calculations.

constituted board of directors and financial reporting. This explains why the tables and figures that follow are updated to end-year 2017.

Figure 9.2 shows the breakdown of Barclays' asset side of the balance sheet in 2017. About 35 per cent of the bank's assets derive from loans to retail and corporate customers (compared to 64 per cent at end-year 2004, according to Barclays' 2004 *Annual Report*). Specifically, these items include home loans, credit cards, unsecured and other retail lending (about 40 per cent of the item total loans and advances to customers) and corporate loans (60 per cent). Derivative financial instruments represent the second largest item in Barclays' assets

Table 9.4 Barclays Bank assets, 2013–2017 (£mil)

Assets	2013	2014	2015	2016	2017
Cash and balances at central banks	42,139	35,469	43,669	97,466	165,713
Items in course of collection from other banks	992	801	596	1,010	1,011
Trading portfolio assets	66,212	49,076	33,057	36,527	79,836
Financial assets designated at fair value	80,621	40,867	71,987	80,865	117,182
Derivative financial instruments	345,434	426,565	316,252	327,202	232,288
Financial investments	82,272	78,590	83,608	61,599	54,583
Loans and advances to banks	51,650	57,438	62,201	36,391	37,255
Loans and advances to customers	462,583	429,814	395,207	423,124	388,960
Reverse repurchase agreements and other similar secured lending	142,695	102,824	28,803	22,941	22,964
Prepayments, accrued income and other assets	16,685	16,529	9,103	4,730	3,429
Investments in associates and joint ventures	182	193	127	164	165
Investments in subsidiaries	14,374	18,345	17,782	14,399	14,614
Property, plant and equipment	1,724	1,601	1,502	1,454	565
Goodwill and intangible assets	4,806	4,945	5,340	5,651	3,498
Current and deferred tax assets	84	2,079	2,096	2,764	1,978
Retirement benefit assets	2,736	–	807	7	959
Assets included as disposal groups classified as held for sale	–	620	5,180	3,453	–
Total assets	**1,315,189**	**1,265,756**	**1,077,317**	**1,119,747**	**1,125,000**

Source: **www.investorrelations.barclays.co.uk**

side of the balance sheet (21 per cent of total assets). Approximately 10 per cent of total assets is represented by financial assets designated at fair value. It is useful to note that Barclays switched to the new financial reporting standards that were approved by the EU Commission with effect from 1 January 2005 and are compulsory for all listed companies. Accordingly, the valuation of some assets and liabilities and the format of the income statements were subject to significant modifications. For instance, trading and financial assets and liabilities are now designated at **fair value**. Box 9.1 explains the meaning of adopting fair value compared to book value.

Tables 9.4 and 9.5 show that the total size of the bank's balance sheet at end-year 2017 is over £1.1 trillion; over 2013–2017 the size of the balance sheet has decreased by 14.5 per cent. This has been brought about mainly by a reduction in derivatives (−33 per cent) and loans (−16 per cent).

On the liability side, customer deposits account for nearly 42 per cent of the total. Only 5 per cent of total liabilities are represented by short- and long-term debt securities (e.g. commercial paper, CDs and bonds). As for the asset side, the relative importance of derivative financial instruments is again large (around 21 per cent). Table 9.5 also shows details on shareholders' equity (see Chapter 7 on bank capital regulation).

Table 9.5 Barclays Bank liabilities, 2003–2017 (£mil)

Liabilities	2013	2014	2015	2016	2017
Deposits from banks	64,667	70,342	63,682	50,464	38,364
Items in the course of collection due to other banks	1,168	911	758	636	446
Customer accounts	497,259	425,134	411,791	431,001	448,079
Repurchase agreements and cash collateral on securities lent	148,545	95,373	26,164	29,226	49,883
Trading portfolio liabilities	28,990	25,910	23,567	31,999	41,542
Financial liabilities designated at fair value	77,926	62,716	90,198	101,330	169,044
Derivative financial instruments	335,223	419,605	310,025	317,874	229,227
Debt securities in issue	62,812	63,771	45,720	60,864	55,874
Subordinated liabilities	20,982	20,851	21,411	23,878	24,203
Accruals, deferred income and other liabilities	19,042	18,649	17,779	7,853	6,885
Provisions	3,313	3,790	3,681	3,515	3,028
Current tax liabilities	696	408	701	767	242
Retirement benefit liabilities	1,588	1,309	213	207	149
Liabilities included in disposal groups classified as held for sale	–	275	4,103	2,135	–
Total liabilities	**1,262,211**	**1,209,044**	**1,019,793**	**1,061,749**	**1,066,966**
Shareholders' equity					
Shareholders' equity excluding non-controlling interests	52,978	56,712	57,524	57,998	58,034
Non-controlling interests	–	–	–	–	–
Total shareholders' equity	52,978	56,712	57,524	57,998	58,034
Total liabilities and shareholders' equity	**1,315,189**	**1,265,756**	**1,077,317**	**1,119,747**	**1,125,000**

Source: www.investorrelations.barclays.co.uk

BOX 9.1 ASSETS AND LIABILITIES VALUATION IN BANKING

Fair value is an accounting term that indicates how a bank values certain assets and liabilities. Under fair value these are stated in the bank's accounts at the current market values and if an active market is unavailable the bank will use estimates. Historical cost accounting is the main alternative to fair value and it refers to the original purchase price paid by the bank to acquire an asset (or received for a liability). However, historical cost also implies that the amount recorded in the financial statements should not exceed the amount expected to be recovered from either its use or its sale.

To understand these principles, it is useful to provide a simple example. Take Bank ABC, which in 2020 buys 1000 shares in a listed bank for €12 per share.

Table 9.6 Bank ABC: shares fair value vs historical cost

Balance sheet	Shares' fair value	Shares' historical cost
End-year 2019	€20,000	€12,000
End-year 2020	€5,000	€5,000

Source: Adapted from ICAEW.

BOX 9.1 Assets and liabilities valuation in banking (*continued*)

Assume that at the end of the year the price per share has increased to €20 and that at the end of 2019, due to a sharp recession, the price drops to €5. Under the two accounting principles, Bank ABC would report the figures in its balance sheet shown in Table 9.6.

The recent shift towards the use of fair value accounting in banking has been driven by several factors including a greater involvement in capital markets activity, rapid financial innovation and the embrace of market-based risk management. Regulatory developments have been led by the US Financial Accounting Standards Board (FASB) and the International Accounting Standards Board (IASB), responsible for two distinct sets of accounting standards: the US GAAP and the IAS/IFRS, respectively. Under both the International Financial Reporting Standards (IAS39) and the US Generally Accepted Accounting Principles (FAS159), banks held a range of financial instruments valued using either fair values or historical value. Assets that banks intend to hold to maturity are valued at historic cost. In contrast, assets for sale or trading assets are valued at market prices (fair value). Any gains or losses relative to sale or trading assets are reported in accumulated other comprehensive income. Fair values are established using level 1 inputs (observable prices in active markets); level 2 inputs (using prices for similar assets in active or inactive markets); and level 3 inputs (using modelling assumptions based on assumed prices, otherwise referred to as 'mark-to-model').

As shown in the table above, in contrast to historic cost, fair value reporting allows increases in value above cost. Vice versa losses can be overstated when values fall. This implies that one of the major drawbacks often attributed to fair value accounting principles is that financial statements become pro-cyclical as they follow market swings. This is particularly relevant for banks compared to other industries because they hold proportionally higher amounts of financial assets and liabilities on their balance sheets.

(For more information, see The Institute of Chartered Accountants in England and Wales (ICAEW), **www.icaew.com**.)

9.2.1.3 Bank equity capital

Defined as the value of assets minus the value of liabilities, the capital (or 'net worth' or 'equity capital') represents the ownership interest in a firm:

$$\text{Capital} = \text{Assets} - \text{Liabilities}$$

Bank capital and liabilities represent the specific sources of funds (see Figure 9.3). However, compared to manufacturing firms, typically banks are highly **leveraged** and thus hold a lower proportion of equity to assets (see Box 9.2). If a relatively small amount of loans are not repaid, this can seriously affect the level of equity and leave the bank technically insolvent. This is because if loans are not repaid then losses have to be borne by the capital cushion that banks hold to protect against such losses. The greater the level of capital relative to the losses incurred, then the greater protection the bank will have. If losses exceed the level of capital then a bank will become technically insolvent because even if it could liquidate all its assets there would not be sufficient funds to cover deposits. In such circumstances, the need to ensure depositors' confidence (a major issue for the banking sector) may result in one of the following:

1. other banks can engage in a rescue package to pump new capital into the troubled bank; or
2. the authorities can decide to rescue the troubled bank using taxpayers' money. The potential repercussions on the whole banking sector are such that regulatory authorities monitor bank behaviour and try to ensure that banks have adequate capital and that they are run in a safe and sound manner (see also Section 7.7).[2]

[2] Adequate capital corresponds to the 'C' in the CAMELS structure that includes also Asset quality, Management quality, Earnings, Liquidity and Sensitivity to market risk. For more details see Section 8.3.2.

In general, the primary function of capital is to reduce the risk of failure by providing protection against operating and any other losses. It does this in five ways by:

1 providing a cushion for firms to absorb unanticipated losses with enough margin to inspire confidence and enable the bank to remain solvent;
2 protecting uninsured depositors (depositors not protected by a deposit insurance scheme that covers small depositors) in the event of insolvency and liquidation;
3 protecting bank insurance funds and taxpayers;
4 providing ready access to financial markets and thus guarding against liquidity problems caused by deposit outflows; and
5 limiting risk taking.

Capital is also needed to acquire plant and other real investments that are necessary to provide financial services. For example, a bank will need capital for its technological investments, branching network and for the management of the payment systems. A bank can also use its capital resources to finance acquisitions.

Capital and risk are strictly connected. Generally speaking, more risk requires more capital so capital adequacy should be a function of risk exposure, all other things being equal. Today banks are exposed to many different financial risks; this is because their activities are increasingly taking place in markets that can be affected by changes in interest and exchange rates as well as variations in credit conditions that can affect both on- and off-balance-sheet

BOX 9.2 TYPICAL CAPITAL STRUCTURE OF A MANUFACTURING FIRM VERSUS A RETAIL BANK

Table 9.7 illustrates the typical balance sheet composition of a bank versus a manufacturing firm. Banks have a higher proportion of short-term assets over total assets and a smaller proportion of fixed assets. On the liability side, banks, on average, have a higher proportion of short-term liabilities (deposits). One major difference relates to the amount of leverage. Manufacturing firms hold a much higher proportion of capital to asset compared to banks.

Table 9.7 Banks vs. manufacturing firms – balance sheet composition

Manufacturing firm	%	Bank	%
Assets		**Assets**	
Short-term assets	55	Short-term assets	70
Fixed assets	45	Long-term and fixed assets	30
Total assets	100	Total assets	100
Liabilities		**Liabilities**	
Short-term liabilities	25	Short-term liabilities	80
Long-term debt	35	Long-term debt	12
Shareholders' equity	40	Shareholders' equity	8
Total liabilities	100	Total liabilities	100

BOX 9.2 Typical capital structure of a manufacturing firm versus a retail bank (*continued*)

The value of the manufacturing firm would have to decline by more than 40 per cent before the firms would become insolvent while a decline of only 8 per cent would make the bank insolvent.

The debt/equity ratio (or financial leverage) of the manufacturing firm is 60/40 = 1.5 and the debt/equity ratio of the bank is 92/8 = 11.5.

The structure of the balance sheet is extremely important for all firms. It is obvious that the way it is leveraged affects the value of the firm; it is an objective for financial managers to achieve a level of debt/equity that maximises the value of the company.

Source: Adapted from Koch and MacDonald (2014), p. 458.

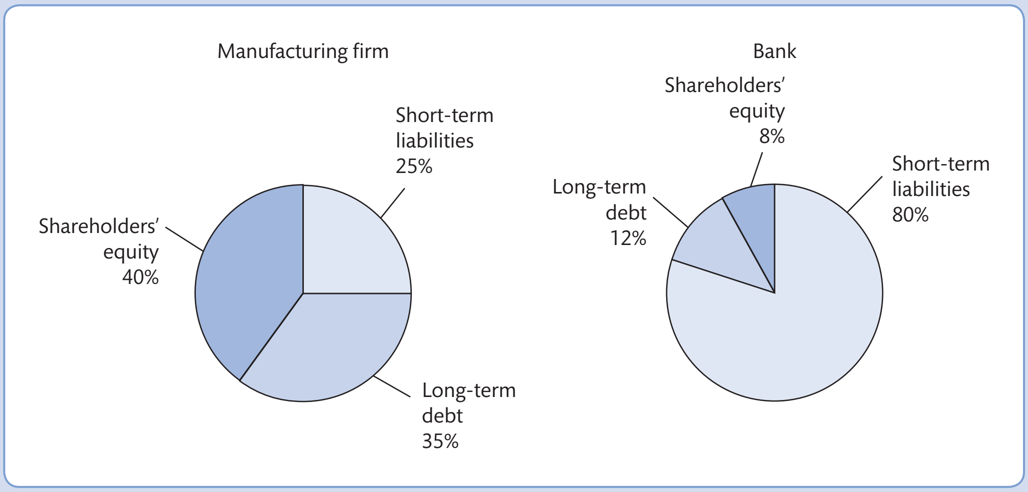

Figure 9.4 Banks vs. manufacturing firms – balance sheet composition

positions. In such a context banks' need for capital is much higher compared to the past. The recent turmoil in the global marketplace has prompted regulatory authorities and policy makers to revise the capital rules so that banks' financial statements reflect more adequately the level of risks undertaken (see Section 7.7).

9.2.1.4 Banks' income structure

The profitability of a bank can be derived from its income statement. Also known as its **profit and loss account** (or *Report of Income* in the US), this measures bank performance between two year-end balance sheets. The relationship between the balance sheet and income statement relates to the fact that the balance sheet reports stock values (e.g. the amount of outstanding loans), whereas the income statement represents cash flow values for a particular year (e.g. the interest received on outstanding loans). Therefore, the income statement reflects the revenue sources in banking as well as the costs.

The **costs**, derived from the liabilities side of the balance sheet, relate to the payments that banks have to undertake such as payment of interest on deposits, dividends to shareholders, interest on debt, provision for loan losses and taxes. The **revenues**, generated by the assets, include interest earned on loans and investments, and fees and commissions (interest and non-interest revenue). Then as any other firm, banks also incur staffing and other operating costs.

$$\text{Bank profits} = \text{Income} - \text{Costs}$$

In the relationship above, income is equal to interest and non-interest income; costs are the sum of interest costs, staff costs and other operating costs. Table 9.8 shows a simplified income statement and how **profits** are calculated for a retail bank.

The *interest income* is the income generated on all banks' assets, such as loans, securities and deposits lent out to other institutions, households and other borrowers. *Interest expense* is the sum of interest paid on all interest-bearing liabilities such as all deposit accounts, CDs, short-term borrowing and long-term debt. The difference between interest income (revenues) and interest expenses (costs) is the *net interest income* (NII).

Provision for loan losses (PLL) is the amount charged against earnings to establish a reserve sufficient to absorb expected loan losses. It can be subtracted from net interest income in recognition that some of the reported interest income overstates what will actually be received after loan defaults. Thus, *net interest income after provisions for loan losses* is calculated as the difference between NII and PLL.

Non-interest income is the income generated by fee income, commissions and trading income and has become important due to increased emphasis on this source of revenue in recent years. It includes, for example, fees and deposit service charges, such as fees paid on safe-deposit boxes; commissions (e.g. from insurance sales) and gains/losses from trading in securities; and other non-interest income sources such as gains/losses on foreign transactions and from undertaking other OBS activities (such as securities underwriting).

Non-interest expenses include salaries and fringe benefits paid to employees, property and equipment expenses, and other non-interest expenses (such as deposit insurance premiums and depreciation). *Net non-interest income* will be the difference between non-interest income and non-interest expenses.

Table 9.8 A simplified bank income statement

a	Interest income
b	Interest expense
$c\ (= a - b)$	Net interest income (or 'spread')
d	Provision for loan losses (PLL)
$e\ (= c - d)$	Net interest income after PLL
f	Non-interest income
g	Non-interest expense
$h\ (= f - g)$	Net non-interest income
$i\ (= e + h)$	Pre-tax net operating profit
l	Securities gains (losses)
$m\ (= i \pm l)$	Profit before taxes
n	Taxes
o	Extraordinary items (net)
$P\ (= m - n - o)$	Net profit
q	Cash dividends
$r\ (= p - q)$	Retained profit

As we move down Table 9.8 we find the item *pre-tax net operating profit*, which is the sum of interest income minus PLL plus net non-interest income. *Profit before taxes* will be equal to *Pre-tax net operating profit* $\pm$ the *securities gains* (*losses*) that may occur when the bank sells securities from its portfolio at prices above the initial cost to the bank. By deducting taxes and other net extraordinary items (which are unusual or infrequent events that can include, for example, the net revenue from the sale of real assets), it is possible to obtain the *net profit*, which is the profit after tax. Finally, *retained profits* will be equal to net profit minus dividends.

Table 9.9 shows a profit and loss account for Barclays over 2013–2017.

Table 9.9 Barclays Bank PLC profit and loss account, 2013–2017 (£mil)

	2013	2014	2015	2016	2017
Interest income	18,315	14,194	13,953	14,423	13,631
Interest expense	(6,715)	(4,108)	(3,345)	(2,966)	(3,883)
Net interest income	**11,600**	**10,086**	**10,608**	**11,457**	**9,748**
Fee and commission income	10,479	8,622	8,470	8,625	8,775
Fee and commission expense	(1,748)	(1,500)	(1,611)	(1,789)	(1,901)
Net fee and commission income	**8,731**	**7,122**	**6,859**	**6,836**	**6,874**
Net trading income	**6,553**	**3,086**	**3,426**	**2,795**	**3,387**
Net investment income	**680**	**1,309**	**1,097**	**1,324**	**859**
Net premiums from insurance contracts	732	–	–	–	–
Other income	148	160	50	57	69
Total income	**28,444**	**21,763**	**22,040**	**22,469**	**20,937**
Net claims and benefits incurred on insurance contracts	(509)	–	–	–	–
Total income net of insurance claims	**27,935**	**21,763**	**22,040**	**22,469**	**20,937**
Credit impairment charges and other credit provisions	(3,071)	(1,821)	(1,762)	(2,373)	(2,336)
Net operating income	**24,864**	**19,942**	**20,278**	**20,096**	**18,601**
Employment costs	(12,155)	(9,860)	(8,853)	(9,211)	(6,445)
Infrastructure costs	(3,531)	(2,895)	(2,691)	(2,937)	(2,068)
Administration and general expenses	(4,113)	(3,069)	(2,983)	(3,200)	(6,476)
Other expenses	(2,173)	(2,360)	(4,009)	(1,000)	(700)
Operating expenses	**21,972**	**18,184**	**18,536**	**16,348**	**15,689**
Share of post-tax results of associates and joint ventures	(56)	28	41	70	70
Loss/Profit on disposal of subsidiaries, associates and joint ventures	6	(473)	(637)	565	184
Gains on acquisitions	26	–	–	–	–
Profit before tax	**2,868**	**1,313**	**1,146**	**4,383**	**3,166**
Tax	(1,571)	(1,121)	(1,149)	(1,245)	(2,125)
Loss/Profit after tax from continued operations	–	192	3	3,138	1,041
Loss/Profit after tax from discontinued operations	–	653	626	591	(2,195)
Loss/Profit after tax	**1,297**	**845**	**623**	**3,729**	**(1,154)**

At year-end 2017 Barclays Bank had total income, of over £20.9 billion, a decrease by about 26 per cent from 2013. The decrease had been brought about by a sharp drop in fee and commission income (−21 per cent) and net trading income (−48 per cent). This latter was already visible in 2014 and was primarily reflecting lower volatility and subdued trading activity, combined with tighter spreads and fair value reductions.

While the income statement gives a good indication of the profitability of a commercial bank, **bank performance** over time is usually measured in relation to ratio analysis that uses the information contained in both the balance sheet and the income statements. Section 9.4 focuses on the importance of ratio analysis and how to interpret the most common financial ratios.

Before moving on to ratio analysis, Section 9.3 illustrates the main characteristics of investment banks' financial statements and how they compare with commercial banks.

9.3 Investment banks' financial statements

We saw in Chapter 3 that large-scale wholesale financing activities are typically carried out by investment banks. Moreover, investment banks offer a range of services such as securities underwriting (including the issue of commercial paper, Eurobonds and other securities) and provide corporate advisory services on mergers and acquisition (M&As) and other types of corporate restructuring. In a nutshell, investment banks mainly deal with corporations and other large institutions and they typically do not deal with retail customers, apart from the provision of upmarket private banking services. As a result of the global financial crisis of 2007–2009, US investment banks have converted to Bank Holding Companies (BHC) to qualify for government assistance (as did Goldman Sachs and Morgan Stanley), or they have been acquired by commercial banks (as in the Merrill Lynch acquisition by Bank of America). In the remainder of this Section, we use the term investment bank to describe a financial institution that engages mainly in the business typical of investments bank, either as a stand-alone institution or as part of a group. As per 2019, the world largest investment bank (by share of revenues) is JPMorgan Chase, followed by Goldman Sachs. In the UK, one of the most successful investment banks is Barclays, as the bank has a large investment bank division (Barclays Investment Bank, formerly known as Barclays Capital).

To understand in some detail the nature of investment banks' business, it is useful to examine the composition of their statements. This is relatively straightforward when looking at the annual reports of investment banks that have converted to BHCs after the crisis (such as Goldman Sachs, see Box 9.3). It becomes slightly more complicated for banks that have been acquired by BHCs, since their business has been deconstructed across various business segments, as in the case of Merrill Lynch, whose original bank name is only used for the private banking/wealth management of Bank of America, but the investment banking business has been allocated to two business lines: *Global Banking* and *Global Markets*.[3]

[3] In addition to Global Banking and Global Markets, Bank of America currently has another two business segments: Consumer and Business Banking and Global Wealth and Investment Management, with the remaining operations recorded in 'All Other'. The most profitable in terms of net income in 2019 was *Consumer and Business Banking*, which is comprised of deposits to consumers and small businesses, credit and debit card services, and business banking, and offers a diversified range of credit, banking and investment products and services to consumers and businesses.

9.3.1 Investment banks' balance sheet

Table 9.10 exhibits a simplified investment bank balance sheet.

Assets side

On the asset side investment banks keep *cash and other non-earning assets*. These assets include, for example, short-term highly liquid securities along with assets set aside for regulatory purposes.

Another key item is *trading assets*. These are the banks' trading activities that consist primarily of securities brokerage, trading and underwriting, and derivatives dealing and brokerage. Generally, trading assets include cash instruments (e.g. securities) and derivatives instruments used for trading purposes to manage risk exposures. Other cash instruments can include, for instance, loans held for trading purposes (i.e. loans that can be traded in the secondary market).

Investment banks enter into secured lending in order to meet customer needs, and obtain securities for settlement. Under these transactions, they can receive collateral from resale agreements and securities borrowed transactions, customer margin loans and other loans. *Securities financing transactions* are collateralised securities that the bank can sell or re-pledge.

Securities owned for non-trading purposes are classified as *investment securities*. They are marketable investment securities and other financial instruments the bank owns, and can include highly liquid debt securities such as those held for liquidity management purposes, equity securities and other investments such as long-term ones held for strategic purposes. Investment banks' lending and related activities such as loan originations, syndications and securitisations (see Chapter 18) are reported under *loans, notes and mortgages*.

Other investments include other receivables such as amounts due from customers on cash and margin transactions. *Fixed assets* consist of equipment and facilities. Typical examples are technology hardware and software and owned facilities (e.g. premises). *Other assets* consist of intangible assets and goodwill as well as assets generated from any unrealised gains on derivatives used to hedge the bank's borrowing and investing activities. It can also include prepaid expenses and real estate purchased for investment purposes.

Table 9.10 A simplified investment bank balance sheet

Assets	Liabilities
Cash and other non-earning assets	Short-term borrowings
Trading assets	Trading liabilities
Securities financing transactions (receivable)	Collateralised securities
Investment securities	Long-term borrowing
Loans, notes and mortgages	Deposits
Other investments	Other payables
Fixed assets	
Other assets	
	Equity
	Other capital terms
Total assets	**Total liabilities and shareholders' equity**

Liabilities and equity

As shown in Table 9.10, funding of investment banks derives from various sources. The main items are as follows:

- *Collateralised securities* – derived from the bank entering secured borrowing transactions and securities sold under agreement to repurchase; this includes payables under repurchase agreements and payables under securities loaned transactions. (This item corresponds to securities financing transactions on the asset side.)
- *Trading liabilities* – include activities that the investment bank undertakes based on future expectations such as trading securities and derivatives dealing and brokerage.
- *Short-term borrowings* – consists of notes payable and any other short-term borrowings.

The investment bank can issue *other short-term debt instruments* – which may be linked to the performance of equity or other indices – and *medium- and long-term debt instruments*.

Another liability is *deposits* (savings and time deposits), which are typically high-volume corporate deposits, followed by *other liabilities* to customers, brokers and dealers, etc., and finally, *shareholders' equity*.

9.3.2 Investment banks' income statements

Investment banks, like commercial banks, are required to publish their profit and loss accounts (or 'statement of earnings'), which report all costs, revenues and net profits for the financial year. Investment banks' revenues derive from the following four sources:

- trading and principal investments;
- investment banking;
- asset management, portfolio service fees and commissions; and
- interest income.

The first components of *trading and principal investments* relate to income generated from trading in equities and equity derivatives, corporate debt, debt derivatives, mortgage and municipals, government and agency obligations, and foreign exchange. *Principal investments* are those securities held over time for general investment purposes. *Investment banking* (in the US) generally includes underwriting and financial advisory services (e.g. M&As advice). *Asset management and portfolio services* can originate revenues in the form of commissions (e.g. agency transactions for clients on main stock and futures exchanges). More specifically, asset management is a source of fees for investment banks generated by providing investment management (e.g. managing company pension funds and other investments) and advisory services to both individuals and institutions.

Securities services can also generate fees from various activities such as brokerage, financing services and securities lending, and matched book businesses. Finally, *interest income* derives primarily from wholesale lending activity of the bank.

On the cost side, interest expenses can be relatively high due to investment banks obligations on borrowings as well as bond and short-term paper instruments (compared to commercial banks), while the bulk of operating expenses relates to staff costs. Other costs include, among others:

- communication and technology;
- occupancy and related depreciation;

- brokerage, clearing and exchange fees;
- professional fees;
- marketing;
- other expenses.

Box 9.3 illustrates the financial statement composition of Goldman Sachs.

BOX 9.3 GOLDMAN SACHS FINANCIAL STATEMENTS (2019)

Founded in 1869 by a German immigrant, Marcus Goldman, and headquartered in New York, Goldman Sachs today 'is a leading global investment banking, securities and investment management firm that provides a wide range of financial services to a substantial and diversified client base that includes corporations, financial institutions, governments and high-net-worth individuals' (**www.goldmansachs.com**). As of 31 December 2019 the company had 38,300 staff members throughout the world and net revenues of US$36.55 billion. Goldman Sachs provides a variety of services, from capital markets services, investment banking and advisory services, wealth management, asset management, banking and related products and services. In 2008 Goldman Sachs converted into Bank Holding Company and thus is subject to consolidated regulatory capital requirements administered by the Fed.

Figure 9.5 illustrates the assets and liabilities composition for Goldman Sachs in 2019.

On the asset side, relevant items are trading assets at fair value that comprise, for example, securities and financial derivatives held by the bank for trading purposes. These constitute the bulk of total assets (36 per cent). Other relevant items are collateralised agreements, i.e. securities borrowed and securities purchased under agreement to resell, amounting to 22 per cent overall. It is also worth noting that Goldman Sachs holds a relatively high proportion of liquid assets (around 13 per cent).

The assets and liabilities structure of investment banks usually indicates shorter-maturity characteristics on the assets side of the balance sheet compared with a traditional commercial bank. Looking at the liabilities and shareholders' equity side, Goldman Sachs's funding sources derive mainly from deposits

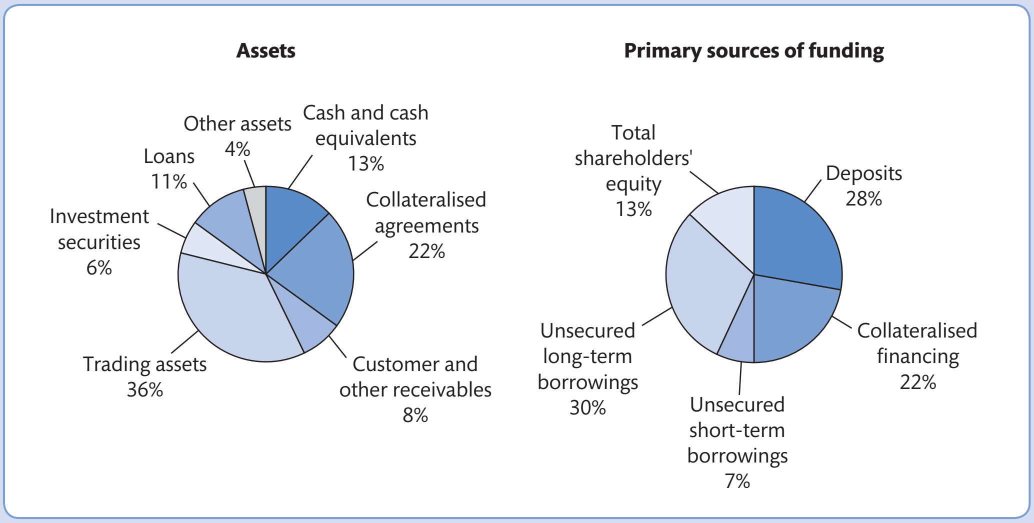

Figure 9.5 Goldman Sachs: asset composition and primary sources of funding, 2019

Source: Goldman Sachs (2019) *Annual Report* and authors' calculations.

BOX 9.3 Goldman Sachs' financial statements (2019) (*continued*)

(28 per cent of total funding sources), long-term borrowing (30 per cent) and collateralised financing (22 per cent).

The revenue sources and cost characteristics of Goldman Sachs for 2019 are shown in Figure 9.6.

The figures show that most revenues derive from market making and other principal transactions (44 per cent), asset management and commissions (25 per cent) and investment banking activities (19 per cent). It is notable that Goldman Sachs earns 40 per cent of its net revenues from its Global Markets business segment and over 60 per cent of its net revenues in the Americas. Operations in Europe, the Middle East and Africa account for 32 per cent (Figure 9.7).

On the operating costs side, staff expenses (in the form of employee compensation and benefits) are prevalent (50 per cent). It is worth noting that on the cost side the proportion of compensation and benefits is about 25:75 of total costs (Figure 9.9).

Source: Goldman Sachs (2019) *Annual Report*.

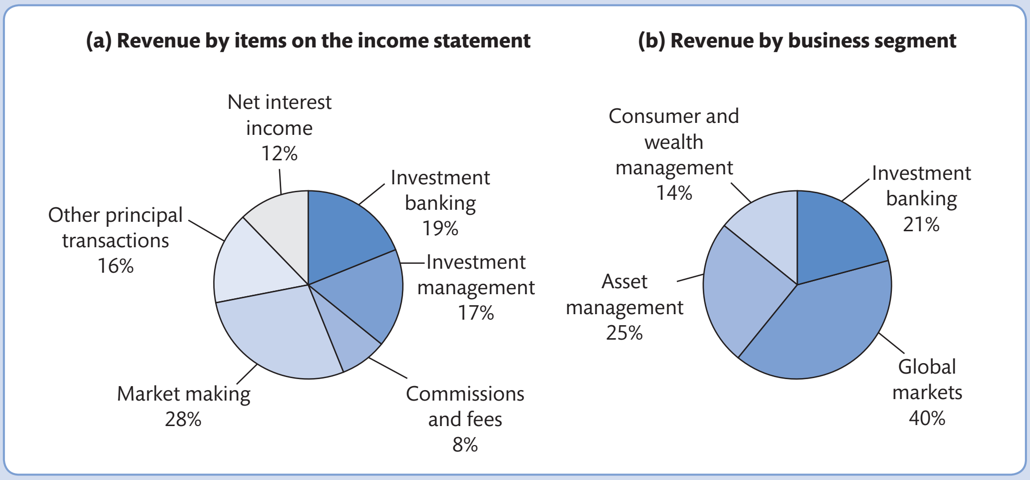

Figure 9.6 Goldman Sachs: sources of revenue, 2019.

Source: Goldman Sachs (2019) *Annual Report* and authors' calculations.

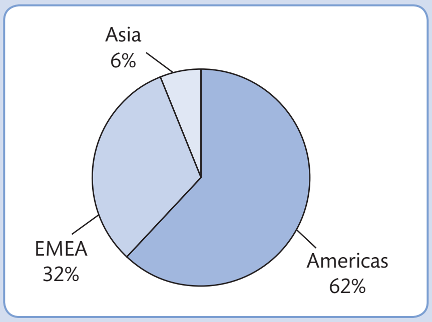

Figure 9.7 Goldman Sachs: net revenues by geographic regions, 2019

Source: Goldman Sachs (2019) *Annual Report* and authors' calculations.

BOX 9.3 Goldman Sachs' financial statements (2019) (*continued*)

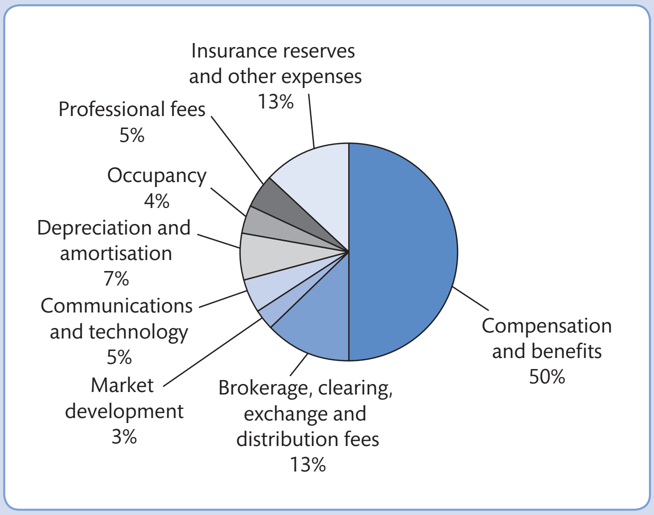

Figure 9.8 **Goldman Sachs: non-interest expenses, 2019**

Source: Goldman Sachs (2019) *Annual Report* and authors' calculations.

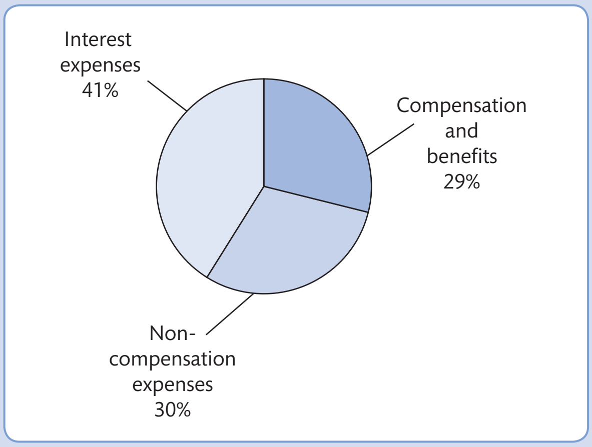

Figure 9.9 **Goldman Sachs: breakdown of costs, 2019**

Source: Goldman Sachs (2019) *Annual Report* and authors' calculations.

9.4 Bank performance and financial ratio analysis

The significant changes that have occurred in the financial sector in all advanced economies have increased the importance of performance analysis for banking institutions. The current operating environment is characterised by more intense competition, greater pressures for banks to control costs, manage risks while at the same time maximise revenues. For publicly listed banks, the objective of shareholders' wealth maximisation (maximising the returns to investors holding equity shares in the bank) is still a priority restrained by increased regulatory constraints (forcing banks to hold more capital reduces returns on capital). It is not surprising that the 2007–2009 financial crisis and the recession that followed have increased the demand for prudential regulation (see Chapter 7). In addition, a broader and more balanced concept of performance has emerged, which is more inclusive and transparent and respects the environment and social challenges. Financial performance analysis is an important tool used by various agents operating either internally to the bank (e.g. managers) or who form part of the bank's external operating environment (e.g. regulators), as shown in Figure 9.10.

- *Shareholders, bondholders*: investors in shares and in bonds issued by the bank, and bank managers and other employees have an obvious economic and strategic interest in the current and future prospects of the banking firm.
- *Direct competitors*: peer group analyses compare the profitability of similar banking institutions operating in similar operating environments; in some cases the homogeneity of the groups being analysed allows for the use of sophisticated statistical techniques.
- *Other market participants*: competitors (or other firms) that represent potential takeover or merger possibilities will rely on **financial ratio analysis** to assess the viability of potential M&A activity and to evaluate potential economic synergies.
- *Financial markets*: capital and money market participants use ratio analysis to monitor the performance of banks. Money market participants, especially those involved with lending in the interbank market will need to assess the credit-worthiness of the banks they are lending to. Deterioration in bank performance may increase credit risk and therefore interbank lenders will require higher returns on their loans. Banks with higher capital ratios will more likely be able to achieve cheaper finance in the interbank markets (as such banks will be perceived as being less risky). Capital market participants and analysts also use ratio analysis to assess the performance of banks as a change in bank performance can alter the valuation of long-term bonds and shares issued by banks. For example, potential bondholders will rely on performance trends as a guide to their investments.

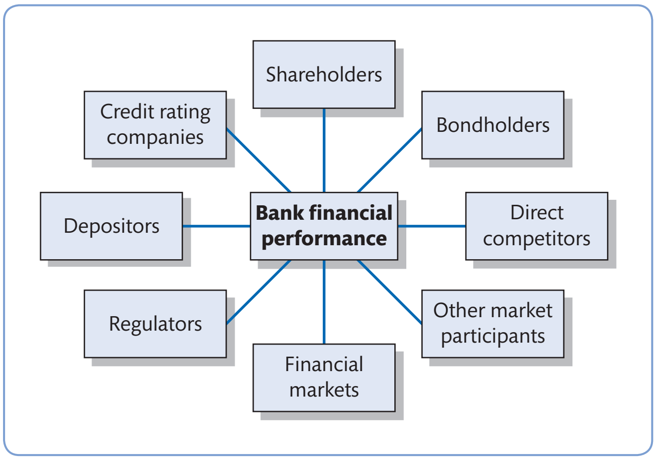

Figure 9.10 Who is interested in bank performance?

- *Regulators*: domestic and international regulatory authorities will also be concerned about the performance of banking institutions. For example, financial regulators need to evaluate the solvency, liquidity and overall performance of banking firms to gauge the likelihood of potential problems. Competition authorities also investigate bank performance indicators to analyse whether banks are making excess profits and behaving in an uncompetitive manner.
- *Depositors*: the smooth performance of banks is valuable for depositors who trust their bank will remain profitable and not expose itself to too much risk.
- Finally, *credit rating companies* – such as Moody's, Standard & Poor's and Fitch IBCA – analyse performance information to compile analyses and ratings of banks operating in a certain country or group of countries.

Bank performance is calculated using ratio analysis and assessed with the aim of (1) looking at past and current trends, and (2) determining future estimates of bank performance. Financial ratio analysis investigates different areas of bank performance, such as profitability, asset quality and solvency.

In addition, over recent years a greater emphasis has been placed on **key performance indicators (KPIs)**. These are 'factors by reference to which the development, performance or position of the business of the company can be measured effectively' (UK Companies Act 2006, Section 417: 6). This definition implies that KPIs can be financial (e.g. cost efficiency, capital adequacy) and non-financial (e.g. customer satisfaction, diversity and sustainability), as described in Box 9.4, and should be monitored and reviewed by management in light of the bank's strategic objectives, operating plan targets and historical performance.

BOX 9.4 WHAT ARE KEY PERFORMANCE INDICATORS (KPIs)?

A report by PwC (2007) stresses the importance of financial and non-financial KPIs for *narrative reporting* to ensure corporate transparency and clarifies their main features and scope, as shown in Table 9.11.

With reference to the specific banking industry, the key strategic drivers rotate around the following six critical aspects: customer retention, customer penetration, capital adequacy, assets under management, asset quality and loan losses.

Recent years have witnessed an increased emphasis on non-financial corporate performance (Schoenmaker and Schramade, 2019). In the European Union,

Table 9.11 Key performance indicators (KPIs)

1 How many KPIs should be reported?	Typically there are likely to be between four and ten measures. However, in order to aid corporate transparency, KPIs should reflect the company-specific business and its strategy so there is no 'one size fits all'.
2 Segmental or group KPIs?	In some cases, it may be more useful to report separate KPIs by business segments (e.g. retail banking, commercial banking, asset management, insurance etc.). For example, in diversified organisations reporting the KPIs only at the group level may be meaningless.
3 How rigid is the choice of KPIs?	Since strategies and objectives evolve over time, it may be appropriate to allow the KPIs to adapt to these changes so there is no requirement to continue reporting the KPIs identified in previous periods.
4 Does reliability matter in choosing KPIs?	Particularly for non-traditional and less well known indicators, it is important that management ensure the reliability and clarity of the information so any limitations should be stated and, if possible, explained to the reader.

BOX 9.4 What are key performance indicators (KPIs)? (*continued*)

all large public interest entities with over 500 employees (listed companies, banks and insurance companies) must comply with Directive 2014/95/EU on non-financial reporting (NFRD), which requires them to disclose in their annual reports details about their commitment to sustainable development. Detailed non-binding guidelines on the disclosure of non-financial information have also been adopted by the European Commission in 2017 (2017/C 215/01) to enhance business transparency on social and environmental matters and ensure consistency and comparability across companies. The guidelines clarify that companies have to disclose 'relevant information on policies, risks, and results as regards environmental matters, social and employee-related aspects, as well as respect for human rights, anti-corruption, and bribery issues, and diversity on the board of directors'. An important role in this context is that of the Global Reporting Initiative or GRI (**www.globalreporting.org/**), which offers support to governments and regulators in the development of sustainability reporting policies and regulations. Sustainability reporting is also central to the so-called 'European Green Deal', which is an action plan of the European Commission to advance sustainable development and tackle climate change so that Europe will be climate neutral by 2050.

Search on the web for the latest non-financial report by Deutsche Bank and answer these questions:

(i) who are the main **stakeholder** groups that the bank has committed to engage with and how does the engagement take place?

(ii) what are the main sustainability ratings used by the bank?

(iii) what is a 'materiality assessment'?

(iv) why are culture and integrity important for a bank?

(v) what can a bank do to protect the environment and human rights when lending?

Source: Adapted from PwC (2007); Schoenmaker and Schramade (2020).

The traditional financial ratios for measuring bank performance are discussed below. The tools that can be used to calculate performance are derived from the information revealed by periodic financial reports produced by the accounting system: the balance sheet and the income statement.

9.4.1 Profitability ratios

Profitability ratios traditionally used in banking are return on equity (ROE), return on assets (ROA), net interest margin (NIM) and C/I (cost-to-income) ratio.

ROA is the return-on-assets calculated as net income/total assets (or alternatively *average* assets over two financial years); this ratio indicates how much net income is generated per £ of assets:

$$\text{ROA} = \text{Net income/Total assets} \tag{9.1}$$

ROE is probably the most important indicator of a bank's profitability and growth potential. It is the rate of return to shareholders or the percentage return on each £ of equity invested in the bank:

$$\text{ROE} = \text{Net income/Total equity} \tag{9.2}$$

Box 9.5 shows the decomposition of ROE into return on assets and equity multiplier. It also summarises briefly the key points of a report by the European Central Bank (2010a) on the shortcomings of this widely used ratio.

BOX 9.5 THE ROE DECOMPOSITION

ROE can be decomposed using a traditional method in corporate finance known as 'Du Pont Model', from the name of the US Corporation that first applied it in the 1920s. This decomposition is important because it allows financial analysts to understand the interrelationship between various ratios and helps banks to invest in areas where the risk-adjusted returns are greater.

We specified in equation (9.2) that accounting ROE is equal to net income divided by total (or, alternative, average) equity. By multiplying ROE by total assets, it is possible to decompose it into two parts: the ROA (= net income/total assets), which measures average profit generated relative to the bank's assets, and the so-called equity multiplier (EM), i.e. a measure of bank's leverage. Specifically:

$$\text{ROE} = \left(\frac{\text{Net income}}{\text{Total assets}}\right) \times \left(\frac{\text{Total assets}}{\text{Total equity}}\right) \tag{9.3}$$

where

$$\text{EM} = \text{Total assets / Total equity} \tag{9.4}$$

so that

$$\text{ROE} = \text{ROA} \times \text{EM} \tag{9.5}$$

ROA can also be split into two parts: **profit margin** (= Net income / Total revenue) and total revenue over total assets, which can be defined as the asset yield or asset 'utilisation' ratio. By multiplying ROA by the bank's total revenue, we obtain:

$$\text{ROA} = \left(\frac{\text{Net income}}{\text{Total revenue}}\right) \times \left(\frac{\text{Total revenue}}{\text{Total assets}}\right) \tag{9.6}$$

Substituting:

$$\text{ROE} = \left(\frac{\text{Net income}}{\text{Total revenue}}\right) \times \left(\frac{\text{Total revenue}}{\text{Total assets}}\right) \times \left(\frac{\text{Total assets}}{\text{Total equity}}\right) \tag{9.7}$$

It is possible to depict the ROE decomposition graphically, as shown in Figure 9.11.

During the 2007–2009 financial crisis, banks that had previously generated high ROE performed rather poorly. An ECB report (2010a) acknowledges the need to better understand the potential trade-off between risk and return in bank performance analysis and discusses the reliability of using ROE as a benchmark particularly in periods of high volatility and weak economic conditions. The major shortcomings of ROE identified by the report can be summarised as follows:

1. There are unbalances in the drivers of ROE (namely ROA and leverage); in particular ROE is not risk-sensitive.
2. Crucial risk elements are missing in ROE such as the proportion of risky assets and the level of solvency.
3. ROE fails to distinguish the best-performing banks from others because it is essentially a short-term indicator and thus cannot measure the potential for *sustainable* (long-term) results of the bank.

BOX 9.5 The ROE decomposition (*continued*)

4 Last but not least, like other accounting metrics ROE may be prone to manipulations from the markets since data are not always reliable and there are significant seasonal factors that can affect them.

Source: European Central Bank (2010a).

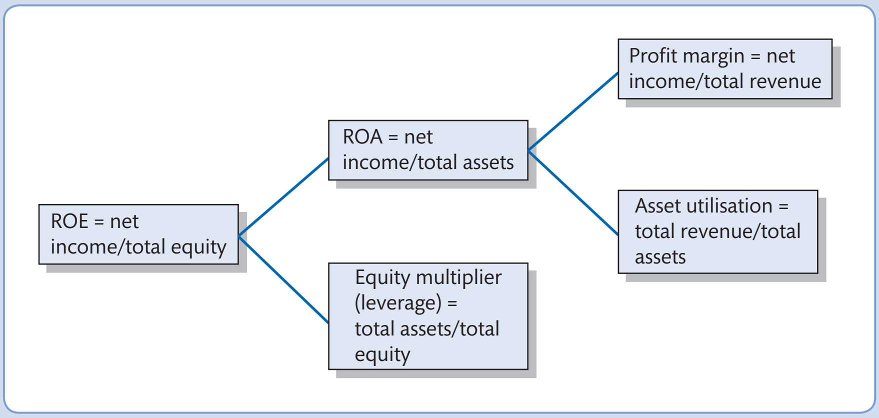

Figure 9.11 The ROE decomposition

Another key performance measure for institutions engaged in traditional banking activities is the NIM, or net interest margin, which measures the net interest income relative to the bank's total, average or earning assets:

$$\text{NIM} = [(\text{Interest income} - \text{Interest expense})/\text{Total earning assets}] \qquad (9.8)$$

NIM reflects the difference between interest earned on assets minus interest costs per £ of assets. It measures the bank's *spread* per £ of assets. A high NIM suggests that the difference between deposit rates and loan (+ other interest earning assets) rates are high, and vice versa. As we have noted in earlier chapters, NIM has been falling in many banking markets reflecting increased competition in the deposits and loan markets – the difference between how much banks pay on deposits and how much they earn on loans is declining.

Finally, the cost-to-income ratio (C/I) is a quick test of efficiency that reflects bank non-interest costs as a proportion of income:

$$\text{C/I} = \text{Non-interest expenses} / (\text{Net interest income} + \text{Non-interest income}) \qquad (9.9)$$

where non-interest expenses are considered as the main inputs to the production process of a bank and total operating income is the output.

If the bank is listed in the capital markets some additional useful measures of performance are EPS (earnings per share), price to book value (P/B) and credit default swaps (CDSs).

$$\text{EPS} = \text{Net income / Average shares outstanding} \tag{9.10}$$

EPS measures the portion of a bank's profit net of tax allocated to each share in issue of common stock. A higher EPS means better profitability for shareholders.

$$\text{P/B} = \text{Stock price / Book value of equity per share} \tag{9.11}$$

Price-to-book value (also known as market-to-book) is the ratio between the market prices of a bank's shares and the accounting value of shareholders' equity per share. The lower the number the better for investors because if, for example, the bank's current share price is £4 and shareholders' equity per share is £2, then investors are ready to pay for one share twice the book value!

Table 9.12 provides the financial highlights for Australia's four major banks: Australia and New Zealand Banking Group Ltd (ANZ), Commonwealth Bank of Australia (CBA), National Australia Bank (NAB) and Westpack Banking Corporation (WBC). As a background, banks' profits suffered a sharp decline due to various structural factors combined with the recession induced by the Covid-19 pandemic. It is possible to note that ROA ranges between 0.25 per cent and 0.75 per cent while ROE ranges between 3.8 and 10.3 per cent (the lowest level in the past 20 years) and NIM around 1.9 per cent.[4] Usually the benchmark for ROA level is around 1 per cent while ROE is considered good when over 10 per cent. High-performing banks typically adopt a target ROE figure of 15 per cent plus. Generally speaking, the higher these ratios the better from a bank's perspective, as higher NIM should

Table 9.12 Australian major banks' (the 'Majors') performance at a glance, 2020

	ANZ	CBA	NAB	WBC
Ranking				
Ranking by total assets	1	2	4	3
Ranking by market capitalisation	4	1	3	2
Performance measures				
Return on assets (%)	0.36	0.75	0.41	0.25
Return on equity (%)	6.2	10.3	6.5	3.8
Net interest margin (basis points)	163	207	177	208
Cost-to-income ratio (%)	52.9	45.9	52.4	61.6
Earnings per share (cents)	132.7	412.5	120.9	72.5
Asset quality				
Impaired loans to total loans	0.40	0.46	0.31	0.40
Capital adequacy ratios				
Tier 1	13.2	13.9	13.2	13.2
Total Capital	16.4	17.5	16.6	16.4

Source: KPMG (2020) and Yahoo finance.

[4] It is not unusual to find ratios expressed in basis points. One basis point is 1/100th or 0.01 per cent. For example, a ratio of 0.055 per cent is 5.5 basis points.

feed through into greater net income, thus boosting ROA and ROE. However, as highlighted in Box 9.5, very high ROE can also indicate high risk taking and unsustainable business models. Table 9.12 also illustrates that CBA is the best-performing bank in terms of cost-to-income ratios. The benchmark for the cost-to-income ratio is around 50–70 per cent, i.e. a low C/I ratio indicates that the bank is operating in an efficient way. Although the major banks in Australia have experienced a drop in profitability, they have continued to strengthen their balance sheets and maintain relatively high capital levels.

Table 9.13 reports the 2019 key profitability ratios for three large former investment banks. It is noticeable that Goldman Sachs' performance is not as strong as that of the other two banks. ROE and ROA exceed 10 per cent and 1 per cent (respectively) for both JPMorgan and Morgan Stanley. Typically, net interest margin has a secondary role for investment banks compared to commercial and retail banks. A more suitable measure of profitability for investment banks is *profit margin*, which is equal to earnings before income taxes to total operating income and takes into account both interest and non-interest income.

Table 9.13 also reports the two market-based measures of performance mentioned above (EPS and P/B). It seems clear that Goldman Sachs is the best bank at generating high EPS while keeping the level of equity in line with the other two banks. The ratio of liquid assets over total deposits and short-term funding appears significantly lower for JPMorgan (74.4 per cent) than its peers.

Finally, spreads on credit default swaps (CDSs) provide useful information, as they represent the cost of insuring an unsecured bond issued by the institution over a specific period of time. CDS spreads are considered a direct indicator of a firm's credit risk (Figure 9.12).

Table 9.13 Selected ratios for three former large US investment banks (%)

End-year 2020	JPMorgan	Goldman Sachs	Morgan Stanley
Return on assets	1.37	0.90	1.06
Return on equity	14.06	9.47	11.26
Net interest margin	2.86	0.85	1.24
Net interest income/Operating revenue	49.56	11.87	16.06
Cost to income ratio	56.99	67.88	72.49
Profit margin[a]	26.03	17.33	21.69
Liquid assets/Deposits	74.41	190.05	198.23
Gross loans/Deposits	63.59	77.26	90.70
Customer loans/Total assets	36.48	14.61	19.24
Impaired loans/Gross customer loans	0.77	3.95	0.85
Loan loss reserves/Impaired loans	171.97	24.92	23.89
Tier 1 ratio	14.15	15.16	18.63
Total capital ratio	16.00	17.81	20.98
Equity/Total assets	9.73	9.27	9.24
EPS (December 2020)[a]	7.67	17.35	5.92
P/B (December 2020)[a]	1.56	1.09	1.34

Note: [a]Data from Yahoo Finance at December 2020.

Source: BankFocus.

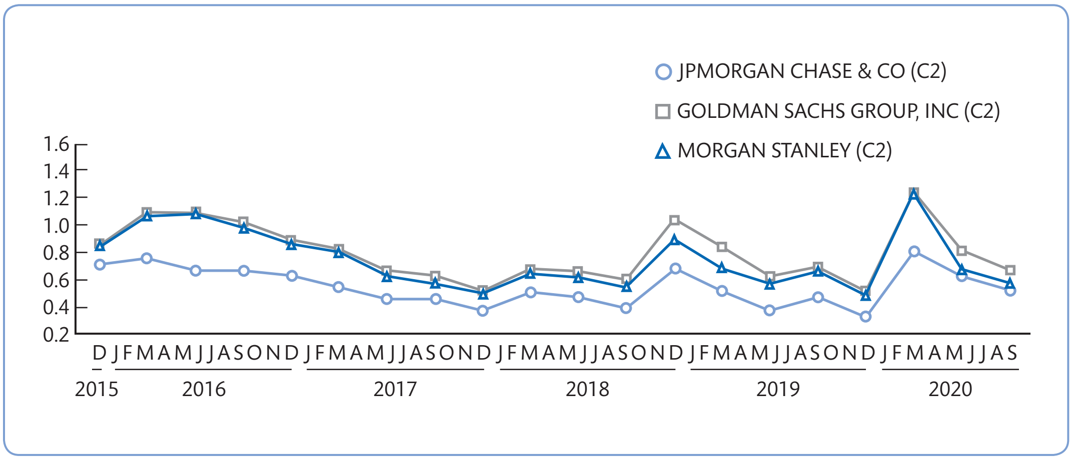

Figure 9.12 CDS spreads evolution for three former large US investment banks (%), 2015–2020
Source: BankFocus.

9.4.2 Asset quality

Lending is still one of the most important activities of banks. While it is expected that all banks will have to bear some positive levels of bad loans and loan losses, one of the key objectives of bank management is to minimise such losses. In the context of the income–expense statement, financial analysts can control the provisions for loan losses (PLL) to manipulate their accounting earnings. For example, more conservative bankers may understate their accounting earnings by building a large and above average loan-loss reserve, while more aggressive bankers may overstate their accounting earnings by keeping the loan-loss reserve low.

The drop in profitability that banks have experienced in recent years has often been due to a poor assets quality that has continued to worsen in a number of European countries as a consequence of the eurozone troubles (see Chapter 14 for a discussion). The European Banking Authority (2019) reports that between 2014 and 2019 in the EU the aggregate Texas ratio dropped by half from 47 per cent to 23 per cent. The Texas ratio is a metrics used for assessing the banks' riskiness and robustness against the problematic assets on the balance sheet. It is calculated as:

$$\text{Texas ratio} = \text{NPLs} / \text{Total tangible common equity capital} + \text{Loan loss reserves} \qquad (9.12)$$

Tangible common equity is the part of shareholders' equity that is not preferred equity and not intangible assets. The higher the Texas ratio, the more serious are the credit problems of the bank. Values >100 imply that problem assets are higher than the resources that the bank may need to cover potential losses on those assets. Figure 9.13 shows the Texas ratio by bank.

The Covid-19 pandemic has posed unprecedented challenges for banks all over the world and has resulted in large banks making substantial provisions for loan losses, as described in Box 9.6.

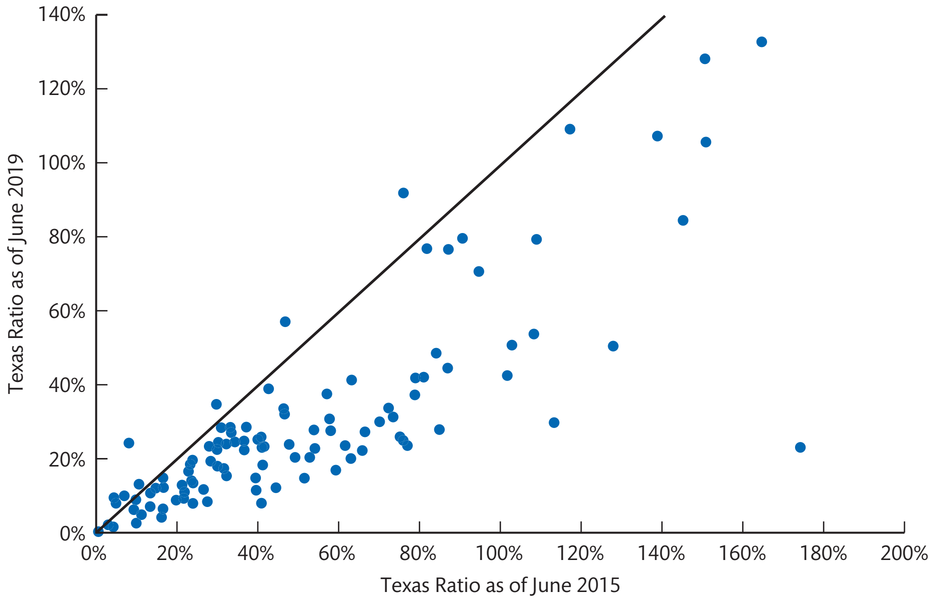

Figure 9.13 Texas ratios by bank, 2015–2019

Source: European Banking Authority (2019).

BOX 9.6 BANKS ACROSS EUROPE BRACED FOR FURTHER HEAVY LOAN LOSS CHARGES

Europe's biggest banks are set to unveil another huge round of provisions for loan losses, as they take stock of the damage wrought by Covid-19 around the globe. The largest UK, Swiss and eurozone lenders are expected to provision at least €23bn for the second quarter as they report earnings in coming days, according to analysts at Citigroup. That is on top of the €25bn of charges the same group took against potential defaults in the first three months of the year. When added to the $61bn already reserved by the five largest US banks over the first six months, the combined figure from the biggest western lenders could reach $117bn. That would be the highest net addition to reserves since the first half of 2009, in the aftermath of the collapse of Lehman Brothers, according to Citi. Few economists are projecting rapid "V-shaped" recoveries and more pain is expected when government support schemes wind down in the autumn. Oliver Wyman, the consulting firm, forecasts as much as €800bn of loan losses for European banks over the next three years if there is a second wave of infections.

"It's going to be another difficult one — several banks have flagged this could be the worst quarter of the year," said Jon Peace, an analyst at Credit Suisse. He noted that under new accounting rules, banks are required to "front-load" their provisions for likely losses, but added that at the end of the first quarter they were working off assumptions for GDP growth and employment "that were not as bleak as today". UBS, the first of the major European financial institutions to report last week, posted a 43 per cent surge in earnings at its investment banking arm, but also took another $272m of loan-loss charges. That brought its total for the first half to $540m — 16 times the same period in 2019. "The first

BOX 9.6 Banks across Europe braced for further heavy loan loss charges (*continued*)

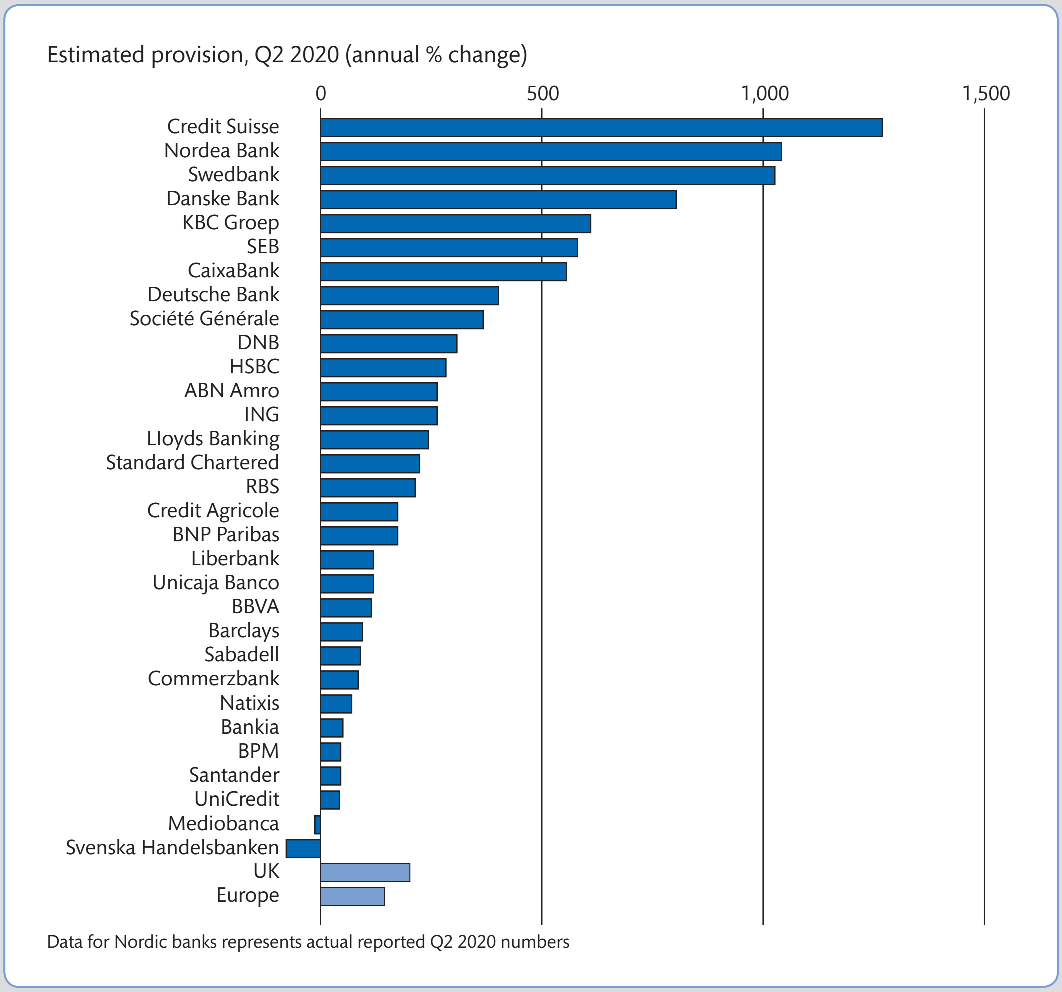

Figure 9.14 Investors braced for a rise in European bank loan losses

Source: Citigroup.

quarter was about whether you are resilient and, for some, able to survive," Sergio Ermotti, chief executive of UBS, told the FT. "The second quarter will be about whether you can demonstrate adaptation. We have already entered the 'lessons learned' phase of coronavirus." The European banking sector, still dogged by problems carried over from the financial crisis of 2007–09, has been punished in the stock market. Shares in European banks have dropped 31 per cent this year, on average, compared with a 10 per cent fall in the benchmark Stoxx 600 index. On average, the banks are trading at less than 40 per cent of the book value of their net assets. Barclays (€22bn), Deutsche Bank (€17bn) and Italy's UniCredit (€20bn) have a combined market capitalisation less than Zoom, the $74bn (€64bn) videoconferencing company that has prospered during the pandemic. "It is a universal consensus that given the headwinds, investing in banks is as stupid an activity as investing in oil majors," said Richard Buxton, head of UK Alpha strategy at Jupiter

BOX 9.6 Banks across Europe braced for further heavy loan loss charges (*continued*)

Asset Management. "It is unlikely that anything revelatory emerges from this reporting season to change that." "The economic downturn clearly means a big pick-up in bad debts," he added. However, "I am extremely confident that whatever the damage to [profit and loss accounts] from the crisis, it does not mean they need to raise additional equity capital."

In the first quarter there was a wide disparity between banks' accounting approaches to potential loan losses — a discrepancy described as "extraordinary" by Magdalena Stoklosa, head of European financials research at Morgan Stanley. An outlier so far is Deutsche, which provisioned just €500m in the first quarter, compared with £2.1bn at Barclays and $3bn at HSBC. The German bank has already said loan-loss provisions would rise to €800m in the second quarter, the highest level in a decade. For European lenders that have significant investment banking arms, a surge in trading revenues should cushion the blow. US banks reported an average 69 per cent increase in revenues from trading stocks, bonds and other assets, benefiting from market volatility and a slew of emergency fundraisings from big companies. Most of those gains came from fixed income, where revenues more than doubled at JPMorgan, Goldman Sachs and Morgan Stanley. While European banks on the whole have smaller debt-trading businesses, Barclays, Deutsche, Credit Suisse and BNP Paribas are positioned to profit the most. On average, analysts expect them to post trading revenues up between 40 to 50 per cent.

The spoils will not be spread evenly, according to Berenberg's Eoin Mullany. While Barclays and UBS have increased their share of global revenues in the past year, "by contrast, the loss of European banks' market share came almost exclusively from Deutsche and Société Générale," he said. Investors are also awaiting the results of the European Central Bank's Covid vulnerability exercise on Tuesday, and an associated decision on whether banks can resume the payment

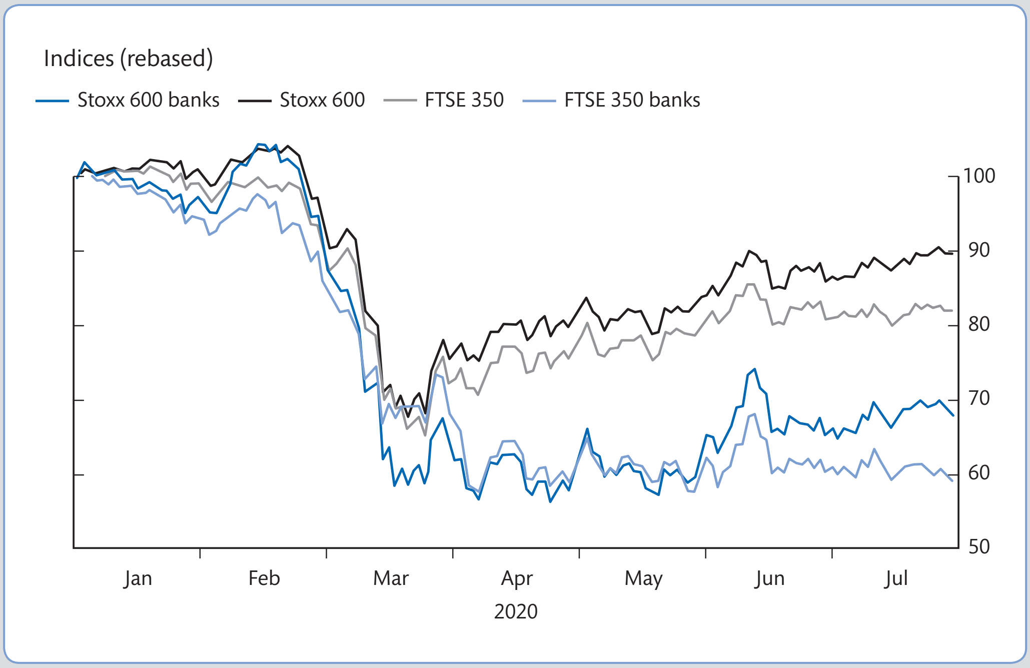

Figure 9.15 Bank stocks are underperforming the wider market

Source: Refinitiv.

BOX 9.6 Banks across Europe braced for further heavy loan loss charges (*continued*)

of dividends. European, Swiss and UK regulators banned payouts at the start of the crisis in mid-March to force banks to conserve capital and lending capacity. Stuart Graham, founder of Autonomous Research, said few investors were expecting the ECB to allow banks to return to normal payouts later this year, saying that January 2021 "is a more realistic date". However, clarity from the regulator will mean fund managers "know the rules of the game once again", he added. "It could attract back those investors for whom the sector had been rendered borderline uninvestable because of the zeroing of dividends."

FT *Source*: Morris, S. and Walter, O. (2020) 'Banks across Europe braced for further heavy loan loss charges', *Financial Times*, 26 July.

9.4.3 Cost of capital and shareholder value creation in banking

While the above discussion of bank performance focuses on traditional and market-based measures, one important indicator that is now widely used by banks (and other companies) relates to what is known as 'shareholder value creation'. The main strategic objective of a profit-oriented bank is to generate *value* for its owners (shareholders) – see Box 9.7.

A bank can create **shareholder value** by pursuing a strategy that maximises the return on capital invested relative to the (opportunity) **cost of capital** (the cost of keeping equity shareholders and bondholders happy). In other words, if a bank invests in a project that generates greater returns than the cost to shareholders of financing the project then this should boost

BOX 9.7 VALUE MAXIMISATION FOR THE BANKING FIRM

As with any other commercial firms listed in the stock market, one of the main objectives of a banking firm is to maximise its value. This is because if the bank meets investors' expectations then it will be able to raise the capital it needs to sustain its future growth. It is well known that the value of the stock (V_0) for any firm will be equal to the present value of expected future stream of dividends $E(D)$ (see Appendix A1 on the concept of present value):

$$V_0 = \frac{E(D_t)}{(1+r)^t} + \frac{E(D_1)}{(1+r)^1} + \ldots + \frac{E(D_2)}{(1+r)^2} = \sum_{t=1}^{\infty} \frac{E(D_\infty)}{(1+r)^\infty} = = \sum_{t=1}^{\infty} \frac{E(D_t)}{(1+r)^t} \quad (9.13)$$

where r is the minimum acceptable – or required – rate of return on the stock, given the bank's riskiness and the returns available on other investments. The rate of return r is the bank's *cost of capital*.

The valuation model above is the well-known dividend discount model, which assumes that the amount of dividends paid in each period varies over time. It also implies that we need dividend forecasts for every year into the indefinite future. By assuming that dividends are trending upward at a stable growth rate that we call 'g', the model can be simplified as follows:

$$V_0 = \frac{D_0(1+g)}{r-g} = \frac{D_1}{r-g} \quad (9.14)$$

BOX 9.7 Value maximisation for the banking firm (*continued*)

This is known as the 'Gordon' model and essentially implies that a bank's stock is expected to grow at the same rate as dividends. For example, suppose the most recently paid dividend for Bank Delta was $D_0 = £5$, $g = 0.04$ and the appropriate discount rate is 12 per cent. By applying the formula, Bank Delta's current share value will be:

$$V_0 = £5 / (0.12 - 0.05) = £71 \text{ per share}$$

returns to holders of the banks' shares (in terms of capital appreciation of stock and higher dividends). The concept can be applied to an individual project, like a bank considering making a strategic investment in another country, or for the whole banks' performance overall.

So shareholder value is created when:

$$\text{Return on capital invested in the project} > \text{Cost of capital to the firm}$$

or

$$\text{Return on capital (ROC)} > \text{Cost of capital}$$

In order to add shareholder value, firms must invest in projects that generate returns exceeding their cost of capital. A common measure of shareholder value is economic value added, as described in Box 9.8.

To calculate the cost of capital we can use the capital asset pricing model (CAPM), where:

$$R_i = R_f + \beta (R_m - R_f) \tag{9.15}$$

where:

R_i is the required rate of return on an investment

R_f is the risk-free rate

R_m refers to the market return

β is a measure of the volatility of the company's equity relative to the overall market.

The CAPM (see Appendix A2) states that investors require a return from holding a company's shares that exceeds the risk-free rate (R_f), to compensate them for holding equity over bonds (this is $R_m - R_f$, otherwise known as the equity risk premium) and for the riskiness of the company relative to the whole market (β).[5]

For example, if a company has a beta (β) of 1.5, and assuming a risk-free rate of 6 per cent (given by the US long-bond rate) and an equity premium of 5 per cent ($R_m - R_f$), then the cost of capital to the firm will be 13.5 per cent. In other words, to maintain shareholder value, this firm will have to invest in projects that generate returns greater than 13.5 per cent if they are to add to shareholder wealth. Investments that generate returns of less than 13.5 per cent will destroy shareholder value. The equity market premium (the difference between equity and bond returns) is usually calculated over a 20- or 25-year period and there is much debate as to how large this premium is, although the US equity market premiums are almost always found to be greater than those in the UK, and they are even lower in continental

[5] A beta of less than one indicates lower risk than the market; a beta of more than one indicates higher risk than the market.

BOX 9.8 WHAT IS EVA?

Bank shareholders rely on performance measures to evaluate how well they are doing in relation to their wealth-maximising objectives. However, profit-based measures ignore the cost of equity capital and are calculated based on accounting standards that do not reflect truly the amount of wealth created. Economic Value Added (EVA) is a performance measurement system that overcomes these two limitations.

EVA was developed by the US consulting firm Stern Stewart & Co and is now commonly used both by practitioners and academics to investigate shareholder value. To increase shareholder value the financial firm has to produce at least a positive equity spread which can be converted into EVA. Specifically, EVA expresses the surplus created by a banking firm (or division within a bank) in a given period, i.e. the firm's profit net of the cost of all capital. It is always a useful measure to calculate the value creation in a specific division or business segment within the bank.

Following Fiordelisi (2007), EVA can be calculated for each bank over the period t and $t-1$ using a procedure accounting for bank peculiarities, specifically:

$$\text{EVA}_{(t-1,\,t)} = \text{NOPAT}_{t-1,\,t} - (\text{CI}_{t-1} {*} K^{e}_{t-1,\,t}) \tag{9.16}$$

where NOPAT is net operating profits after tax, CI = bank's capital invested and K^e = shareholders' expected rate of return.

The calculation of EVA requires that NOPAT and capital invested (CI) are expressed on an economic (rather than accounting) basis. Normally, adjustments are carried out on accounting data, particularly relating to loan loss provision and loss reserves, R&D and training costs, taxes and so on.

Concerning capital invested (CI) and its cost, various studies suggest the use of equity capital, while for the cost of capital (I) it is possible to employ the book value of shareholder equity. Finally, the cost of equity is estimated using the capital asset pricing model (CAPM) and looking at investors' expected returns.

A bank pursuing a focused EVA strategy will have the following priorities:

1 raise the rate of return on assets via revenue-enhancing and cost containment (e.g. branch closures) strategies;
2 the development of capital-free business to raise the measured ROE (e.g. increasing in non-interest fee incomes that can be generated with low capital expenditure);
3 the removal of assets from the balance sheet if they do not meet the target ROE established as the bank's objective (e.g. via securitisation);
4 being prepared, if necessary, to radically change organisational structure (e.g. by outsourcing part of the business);
5 exiting business areas which do not generate a rate of return equal to the target set;
6 repaying capital to shareholders (e.g. by repurchasing stocks from its shareholders).

Source: CIMA (2004); Fiordelisi (2007).

Europe. Betas (β) should also be calculated over long periods as short-term estimates may yield unreliable cost of capital estimates.

Box 9.9 explains the calculation of the cost of capital. Cost of capital calculations can be done for the whole bank or divisions/business areas within a bank in order to determine the allocation of capital within the organisation. For example, if a bank's mortgage business is generating returns greater than the cost of capital but credit card activities are making returns less than the cost of capital, the bank should consider dedicating more capital resources to the former and also should think of ways of boosting returns in (or divesting) its credit card business.

Note that this is just the equity cost of capital and we can extend the analysis to include the cost of debt to present what is known as a weighted cost of capital. Also, one should note that there is a variety of other approaches that can be used to calculate the cost of capital (including a wide range of various accounting and other adjustments) and it should

BOX 9.9 CALCULATING THE COST OF EQUITY CAPITAL

CAPM cost of equity capital

The standard model for the cost of equity capital is the CAPM. Despite some drawbacks, this model continues to be the most widely used tool for business decision making. The formula for calculating the cost of equity capital given its risk is as follows: $R_f + \beta$(Equity risk premium), where R_f is the risk-free rate and β is a measure of how much risk the investment will add to a portfolio that looks like the market. If a stock is riskier than the market, it will have a beta greater than one. If a stock has a beta of less than one, the formula assumes it will reduce the risk of a portfolio. Equity risk premium is the return expected from the market above the risk-free rate.

Risk-free rate

Risk-free rates vary from country to country and are dependent on the low-risk government instrument used as a proxy for the risk-free rate. In the following examples we will use the UK 10-year government bond rate, which was 0.72 per cent on 21 April 2021. For US examples we will use 10-year US Treasury, which stands at 1.61 per cent.[6]

Equity risk premium

A wide range of equity premiums can be quoted from the literature, ranging from 3 per cent to 9 per cent. For this analysis, we will use a rate of 5.31 per cent in the UK and 4.72 per cent in the US, as of January 2021.[7]

CAPM beta figures

Beta figures are obtained from Yahoo Finance. The cost of equity capital for the major UK banks for 2021 are shown in Table 9.14.

Table 9.14 Cost of equity capital for RBS, HSBC, Barclays and Lloyds in 2021

	RBS	Barclays
Risk-free rate	0.72%	0.72%
Equity risk premium (%)	5.31%	5.31%
Beta	1.65	1.52
CAPM 'standard' cost of equity capital (%)	**9.48%**	**8.79%**
	HSBC	**Lloyds**
Risk-free rate	0.72%	0.72%
Equity risk premium (%)	5.31%	5.31%
Beta	0.59	1.43
CAPM 'standard' cost of equity capital (%)	**3.85%**	**8.31%**

[6] https://tradingeconomics.com/bonds

[7] http://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/ctryprem.html

BOX 9.9 Calculating the cost of equity capital (*continued*)

Following the same methodology, we repeat the exercise and evaluate the cost of capital for some of the largest US and European banks in 2021. The results of this exercise are reported in Table 9.15.

Table 9.15 Cost of equity capital for selected US and EU banks in 2021

US banks					
	Goldman Sachs	**Morgan Stanley**	**JPMorgan Chase**	**Citigroup**	**Bank of America**
Risk-free rate (T-bill rate)	1.56%	1.56%	1.56%	1.56%	1.56%
Equity risk premium (%)	4.72%	4.72%	4.72%	4.72%	4.72%
Beta	1.50	1.56	1.21	1.95	1.58
CAPM 'standard' cost of equity capital (%)	**8.64%**	**8.92%**	**7.27%**	**10.76%**	**9.02%**
European banks					
	Deutsche Bank	**UBS**	**Santander**	**BNP Paribas**	**Credit Agricole**
Risk-free rate	−0.26%	−0.18%	0.40%	0.00%	0.00%
Equity risk premium (%)	4.72	4.72	3.1	4.3	4.3
Beta	1.40	1.20	1.79	2.05	2.11
CAPM 'standard' cost of equity capital (%)	**6.35%**	**5.48%**	**5.95%**	**8.82%**	**9.07%**

Source: Data on Risk free rates based on 10-year government bonds from https://tradingeconomics.com/bonds and equity premia from http://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/ctryprem.html

be stressed that cost of capital calculations are never definitive – they vary according to the calculation method used.

9.4.4 Solvency ratios

The Basel III Accord (detailed in Chapter 7) requires banks to hold a minimum overall risk-weighted capital ratio of 8 per cent, of which Tier 1 capital should be at least 6 per cent. Total capital adequacy ratio measures Tier 1 capital + Tier 2 capital; this ratio should be at least 8 per cent. The addition of the capital conservation buffer increases the total amount of capital a bank must hold to 10.5 per cent of risk-weighted assets, of which 8.5 per cent must be Tier 1 capital.

The total capital adequacy ratio cannot be calculated simply by looking at the balance sheet of a bank, as the bank has to classify its assets and off-balance-sheet business according to certain risk categories and varying amounts of capital have to be held according to these risks. Recall from Section 7.8, for example, cash has a 0 per cent risk weighting requiring no capital backing, whereas unsecured loans require 8 per cent capital backing. Both Tier 1 and Tier 2 ratios can only be calculated internally by the bank. Banks have the option of publishing these ratios in their annual report. Financial ratios shown in Tables 9.12 and 9.13

illustrate that in 2019 banks in Australia and the US displayed a level of Tier 1 and total capital significantly above the 6 and 8 per cent minimum requirements.

Finally, we can say that, as equity is a cushion against asset malfunction, the simple equity/assets measure (which we can calculate from bank balance sheets) indicates the amount of protection afforded to the bank by the equity they invested in it. It follows that the higher this figure, the more protection there is. However, remember that this is a crude measure of a bank's financial strength because, unlike the Basel Tier 1 and Tier 2 measures, this ratio does not take into account the riskiness of banking business.

9.4.4.1 The trade-off between safety and the return to shareholders

The amount of capital affects the returns to equity holders and ROE is a good measure for shareholders to know how much profit the bank is generating on their equity investments. Indeed, as we discussed in Box 9.5, ROE is related directly with ROA, as follows:

$$\text{ROA} \times \text{EM} = \text{ROE} \tag{9.17}$$

rearranging:

$$\text{EM} = \text{ROE} \times \frac{1}{\text{ROA}} \tag{9.18}$$

substituting:

$$\text{EM} = \frac{\text{Total assets}}{\text{Total equity capital}} \tag{9.19}$$

where EM is the equity multiplier that measures the extent to which a bank's assets are funded with equity relative to debt. To understand the importance of EM, consider two banks, both having total assets (with the same risk features) equal to £50 million and earning a ROA of 1.5 per cent, as shown in Table 9.16.

Table 9.16 illustrates the trade-off between total capital and ROE. In particular, Bank Alpha displays the highest level of total capital and the lowest level of EM and ROE relative to Bank Beta. However, while the shareholders of Bank Beta will be earning twice as much as those of Bank Alpha, it is not necessarily true that Bank Beta is the most desirable for shareholders as Bank Beta is more risky as it has half the amount of capital backing the same amount of risky assets. There is clearly a trade-off between safety and returns to shareholders.

Table 9.16 An illustration of the trade-off between solvency and profitability

Bank	Total assets (a)	Total capital (b)	EM = (a)/(b)	ROA	ROE = EM × ROA
Bank Alpha	£50,000,000	£5,000,000	10	1.5%	15%
Bank Beta	£50,000,000	£2,500,000	20	1.5%	30%

9.4.5 Limitations of financial ratios

Financial ratios have their own limitations. First, generally one year's figures are insufficient to evaluate the performance of banks, and financial analysts typically look at trends to evaluate the ratios and their fluctuations over a timespan of at least five years. Second, precise comparisons between similar banks may be difficult as they often compete in different markets, have varying product features and customer bases, and so on. As such, ratio analysis may be misleading as it is often difficult to compare 'like with like'. Despite these problems, financial analysts often undertake *peer analysis of* similar banks and this involves the creation of peer groups (see also Section 8.3.1 on peer analysis). Third, ratios do not stand in isolation: they are interrelated. For example, poor profitability may affect liquidity and capital ratios. A bank that performs poorly may have to use its liquid assets (if it has an excess of such assets) to fund future lending thus reducing its liquidity ratios. Large losses may be written out of capital thus reducing capital ratios. Another important factor is that ratios relate to a particular point in time and there are seasonal factors that can distort them. Moreover, figures in the financial statements may be 'window-dressed' – that is, made to look *better than they really are* (as was the case mentioned earlier, referring to banks over- or under-provisioning for bad loans). Similarly, financial statements may be manipulated and may not reflect accepted accounting procedures. That is why both domestic and international regulatory authorities have pointed out the need for more transparency, disclosure and uniformity of bank accounts as the markets become increasingly global. For example, at the EU level all listed companies that are required to publish consolidated accounts are also required to prepare their accounts in accordance with adopted International Financial Reporting Standards (IFRS) for accounting periods beginning on or after 1 January 2005. Last but not least, particularly in cross-country analyses, the usefulness of some financial ratios (e.g. those having income net of tax as numerator like ROE and ROA) can be influenced by tax laws that may differ substantially from country to country.

9.5 Conclusion

This chapter examined the main items contained in banks' financial statements and introduced the key financial ratios used by banks to compare performance. It also highlighted the role of bank capital, simply defined as the difference between assets and liabilities. Typically, banks are highly leveraged compared to non-financial firms; therefore capital management techniques are vital to ensure the solvency of banking institutions. In the chapter we also briefly discussed the concept of shareholder value creation and the cost of equity capital.

Furthermore, the analysis of the income statement (or profit and loss) account has shown the various sources of income (interest and non-interest) and cost structure for banks and how to determine the profitability. The chapter also focused on the different activities that investment banks perform and how this is reflected in the structure of their financial statements. We noted that investment banks' balance sheet structure and income statements differ substantially from those of commercial banks.

Many different agents operating either internally or externally to the banks (from managers to regulators and credit-rating companies) will be interested in their performance, thus the last part of this chapter introduced a selection of key ratios used to gauge bank performance, focusing particularly on profitability, asset quality and solvency. There is also a growing emphasis on corporate responsibility and banks need to balance their short-term objectives with the need to create long-term value (Schoenmaker and Schramade, 2019), to satisfy all stakeholders in the transition to a more sustainable economy.

Key terms

Assets
Balance sheet
Bank performance
Capital
Cost of capital
Cost–income ratio
Costs
Fair value
Financial ratio analysis
Income statement (profit and loss account)
Key performance indicators (KPIs)
Leverage
Liabilities
Net interest margin
Non-financial performance
Profit and loss account (income statement)
Profit margin
Profits
Return-on-assets
Return-on-equity
Revenues
Shareholder value
Stakeholder
Stakeholder value

Key reading

European Central Bank (2010) 'Beyond ROE – how to measure bank performance', Appendix to the Report on EU Banking Structures, September.

Fiordelisi, F. (2007) 'Shareholder value efficiency in European banking', *Journal of Banking and Finance*, 31(7), 2151–2171, July.

Schoenmaker, D. and Schramade, W. (2019) 'Investing for long-term value creation', *Journal of Sustainable Finance & Investment*, 9(4), 356–377.

REVISION QUESTIONS AND PROBLEMS

9.1 What is a bank balance sheet? What are the main items in a commercial bank's balance sheet?

9.2 What is equity capital? What are the functions of capital?

9.3 What is a bank income statement?

9.4 What are the main differences between a bank balance sheet and income statement?

9.5 What are the main differences between commercial and investment banks' financial statements?

9.6 Using the information contained in Tables 9.4 and 9.5, calculate Barclays' ROA, ROE, NIM and C/I ratios.

9.7 Download the annual reports of two large banking groups and critically analyse the pages that describe the KPIs. During your seminar class, discuss in groups the KPIs reported by the two banking institutions and try to identify and explain PWC's six critical aspects described in Box 9.4 in light of the major strategies stated in the annual reports of the respective banks.

9.8 Explain how to calculate the cost of equity capital for a bank. Outline the main advantages of this approach to bank performance measurement compared to using standard profitability ratios.

9.9 Explain the trade-off between solvency and profitability.

9.10 What are the main limitations of bank financial ratios?